
Graph Neural Dynamics via Learned Energy and Tangential Flows

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Abstract

We introduce TANGO, a dynamical-systems-inspired framework for graph representation learning that governs node feature evolution through a learned energy landscape and its associated descent dynamics. At the core of our approach is a learnable Lyapunov function over node embeddings, whose gradient defines an energy non-increasing direction that guarantees stability. To enhance flexibility while preserving the benefits of energy-based dynamics, we incorporate a novel tangential component, learned via message passing, that evolves features while maintaining the energy value. This decomposition into orthogonal flows of energy gradient descent and tangential evolution yields a flexible form of graph dynamics, and enables effective signal propagation even in flat or ill-conditioned energy regions, that often appear in graph learning. Our method is designed to help alleviate oversquashing, and is compatible with different graph neural network backbones. Empirically, TANGO achieves strong performance across a diverse set of node and graph classification and regression benchmarks, demonstrating the effectiveness of jointly learned energy functions and tangential flows for graph neural networks.

1. Introduction

Graph Neural Networks (GNNs) have achieved remarkable success in learning representations for graph-structured data (Bronstein et al., 2021), but they face fundamental challenges when scaling depth or modeling long-range interactions, such as vanishing gradients (Arroyo et al., 2025),

over-smoothing (Nt & Maehara, 2019; Cai & Wang, 2020; Rusch et al., 2023), and over-squashing (Alon & Yahav, 2021; Topping et al., 2022; Di Giovanni et al., 2023; Gravina et al., 2023; 2025). To address these issues, recent works have drawn connections between GNNs and dynamical systems or control theory to understand and mitigate these issues (Poli et al., 2019; Chamberlain et al., 2021b; Eliasof et al., 2021; Gravina et al., 2023; Arroyo et al., 2025). For example, treating a GNN as a continuous dynamical system (or *neural ODE*) opens the door to analyzing stability through the lens of diffusion (Chamberlain et al., 2021b), energy conservation (Rusch et al., 2022), antisymmetric dynamics (Gravina et al., 2023), and Hamiltonian flows (Heilig et al., 2025). In parallel, physics-informed neural architectures have shown that embedding physical priors such as energy conservation or dissipation into neural models can dramatically improve stability and interpretability (Bhattoo et al., 2022; Gao et al., 2022; Brandstetter et al., 2022). The common theme in the aforementioned works is the reliance on the existence of *some* energy functional that is minimized or preserved by the GNN parameterization, which is often relatively simple, such as the Dirichlet energy (Rusch et al., 2023). Beyond GNNs, Lyapunov functions and Lyapunov-stable neural ODEs have also been used to guarantee stability of general neural networks (Lawrence et al., 2020; Rodriguez et al., 2022), including models designed for adversarial robustness of image classifiers where the ODE is regularized so that perturbed inputs converge to the same Lyapunov-stable equilibrium point (Kang et al., 2021), which was also studied for graph adversarial robustness in (Zhao et al., 2023). In contrast, in TANGO, we focus on proposing a learnable energy that is utilized with a learned gradient and tangential flow steps, and we use Lyapunov theory to derive the design of a downstream graph learning framework.

At the same time, it is well-established in bioinformatics and computational chemistry that different, and more complex, energy functions are necessary to accurately model various natural processes. For instance, in protein folding, the energy landscape is often rugged and multi-funnel-shaped, reflecting the presence of multiple stable conformations and transition pathways (Wolynes, 2005). Similarly, in computational chemistry, modeling complex chemical reactions and molecular interactions requires sophisticated potential

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energy surfaces (Senn & Thiel, 2009).

Recently, deep learning has seen growing work on *energy-based models* (EBMs), which learn an energy function to model data distributions (e.g., images or molecules), primarily for generative modeling (LeCun et al., 2006; Xie et al., 2016; Du & Mordatch, 2019; Guo et al., 2023). In contrast, we learn a *downstream task-driven energy* whose parameters are optimized through the loss of a downstream task, such as graph or node classification, rather than generative modeling or a dedicated energy loss function.

These insights motivate a fundamental question: *How can we learn a task-driven energy function, and how can it be effectively leveraged within a GNN architecture to guide representation dynamics?* Unlike energy-based generative models, where the energy function encodes data likelihood, our focus is on learning an energy landscape whose evolution corresponds to solving a downstream task, such as node or graph classification. To address these questions, we propose to decompose feature evolution into two orthogonal components: (i) a *gradient descent* direction that minimizes the learned energy, and (ii) a *tangential direction* that evolves along its level sets, preserving energy. This structured decomposition yields a principled framework that promotes stability, enhances interpretability, and is designed to help alleviate issues such as oversquashing.

Our Approach. We introduce TANGO, a framework for constrained graph dynamics that incorporates a learnable Lyapunov energy function into the message-passing process, where the learned energy governs representation updates through two complementary flows: (1) an *energy descent component*, which drives convergence toward task-relevant solutions, and (2) a *tangential, conservative component*, which preserves energy while retaining flexibility by moving along energy level sets. As illustrated in Figure 1, the descent direction (green) lowers the energy, the tangential direction (blue) moves along level sets, and their combination (orange) defines the full update step, enabling effective information propagation with controlled and stable feature dynamics. TANGO’s Lyapunov-inspired analysis guarantees stability in the sense of feature evolution throughout layers rather than claiming state-of-the-art performance, and our empirical studies then assess the impact of the tangential flow.

Main Contributions. Our contributions are as follows:

1. **Lyapunov-inspired Graph Neural Dynamics.** We introduce TANGO, a novel framework for graph representation learning that decomposes feature evolution into energy descent and tangential components, both parameterized by GNNs.
2. **Theoretical Characterization of TANGO.** We prove that, under mild assumptions, TANGO satisfies Lyapunov conditions, ensuring stable dynamics. Additionally, we show that the tangential component enables expressive yet controlled propagation, and we connect this capacity to its ability to mitigate oversquashing empirically via long-range benchmarks.

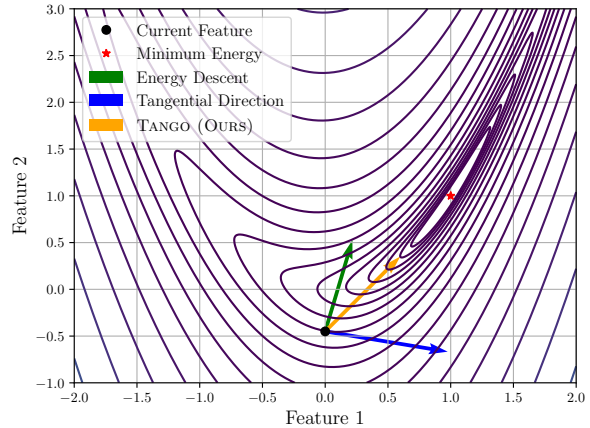


Figure 1. TANGO dynamics in a 2D feature space. We plot the level sets of a learned energy function and visualize the *energy descent direction* (green), the *learned tangential direction* (blue), and their *combined vector* (orange). The tangential component enables movement along level sets, while the descent component reduces energy, allowing an effective navigation of the learned energy landscape.

Lyapunov conditions, ensuring stable dynamics. Additionally, we show that the tangential component enables expressive yet controlled propagation, and we connect this capacity to its ability to mitigate oversquashing empirically via long-range benchmarks.

3. **Strong Empirical Performance.** We evaluate TANGO on a range of graph learning benchmarks, demonstrating performance competitive with or surpassing strong and widely-used baselines.

2. Mathematical Background

In this section, we provide a brief overview of Lyapunov stability theory, based on the classical treatment in Khalil (2002), which underpins the design of our TANGO. This theory originates from control systems and differential equations, offering a principled way to assess whether trajectories of a dynamical system remain bounded and converge over time.

Continuous Dynamical Systems. Let $\mathbf{h}(t) \in \mathbb{R}^d$ denote the state of a dynamical system at time $t \geq 0$, and consider a first-order ODE:

$$\frac{d\mathbf{h}(t)}{dt} = F(\mathbf{h}(t)), \quad (1)$$

where $F : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a continuous vector field. A point $\mathbf{h}^*$ is called an *equilibrium* if $F(\mathbf{h}^*) = 0$.

Definition 2.1 (Lyapunov Function). Let $\mathbf{h}^* \in \mathbb{R}^d$ be an equilibrium of the system in Equation (1). A continuously differentiable function $V : \mathbb{R}^d \rightarrow \mathbb{R}$ is called a *Lyapunov function* around $\mathbf{h}^*$ if:

1. $V(\mathbf{h}) \geq 0$ for all $\mathbf{h}$ in a neighborhood of $\mathbf{h}^*$, and $V(\mathbf{h}^*) = 0$;
2. $\frac{d}{dt}V(\mathbf{h}(t)) = \nabla_{\mathbf{h}}V(\mathbf{h}(t))^\top F(\mathbf{h}(t)) \leq 0$ in that neighborhood.

The first condition ensures that V is lower-bounded by 0, i.e., that the value of the Lyapunov function, sometimes also referred to as *energy*, is non-negative, and the second that V does not increase along trajectories of the system.

We now recall a classical (Khalil, 2002) stability criterion for the dynamical system in Equation (1), based on the definition of a Lyapunov function, which we will later use to characterize the stability of our approach in Section 4.

Theorem 2.2 (Lyapunov Stability). *Let $\mathbf{h}^*$ be an equilibrium of Equation (1) and let V be a Lyapunov function in a neighborhood $\mathcal{N}$ of $\mathbf{h}^*$. If $\frac{d}{dt}V(\mathbf{h}(t)) \leq 0$ in $\mathcal{N}$, then $\mathbf{h}^*$ is Lyapunov stable.*

3. Method

As discussed in Section 1, our goal is to learn a task-driven energy function, and to devise a principled way to utilize it towards improved downstream performance for graph learning tasks, based on a combination of TANGential- and Gradient-steps Optimization of node features. We therefore call our method TANGO. In Section 3.1, we outline the blueprint of TANGO. In Section 3.2, we discuss implementation details. Later, in Section 4, we discuss the properties of our TANGO, and in Appendix C we discuss its complexity.

Notations. We consider a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $n = |\mathcal{V}|$ nodes and $m = |\mathcal{E}|$ edges. Let $\mathbf{H}(t) = [\mathbf{h}_1(t), \mathbf{h}_2(t), \dots, \mathbf{h}_n(t)]^\top \in \mathbb{R}^{n \times d}$ denote the matrix of node features at continuous time t , where $\mathbf{h}_v(t) \in \mathbb{R}^d$ is the state of node v at time t . Following the literature of GNNs based on dynamical systems (Eliasof et al., 2021; Gravina et al., 2023; Arroyo et al., 2025), when considering a discrete architecture with a finite number of layers, we draw an analogy between time t and network depth ℓ . Henceforth, we will interchangeably use the terms $\mathbf{H}(t)$ and $\mathbf{H}^{(\ell)}$ to denote node features at a certain time or layer of the network, depending on the context.

3.1. Optimizing Features with Energy Tangential and Gradient Steps

Our TANGO concept is based on a dynamical system that, given a graph energy function $V_{\mathcal{G}}$, considers two steps: (i) *energy gradient descent* and (ii) *tangential direction* flows, that evolve the node features:

$$\frac{d\mathbf{H}(t)}{dt} = \underbrace{-\alpha_{\mathcal{G}}(\mathbf{H}(t)) \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}(t))}_{\text{Energy Gradient Descent}} + \underbrace{\beta_{\mathcal{G}}(\mathbf{H}(t)) T_{V_{\mathcal{G}}}(\mathbf{H}(t))}_{\text{Tangential Direction}}, \quad (2)$$

where $\alpha_{\mathcal{G}} \geq 0$ and $\beta_{\mathcal{G}} \in \mathbb{R}$ are learned scalars that balance the two steps, $\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}(t))$ is the energy gradient with respect to node features $\mathbf{H}(t)$, and $T_{V_{\mathcal{G}}}(\mathbf{H}(t))$ is an update direction that is orthogonal, i.e., tangential to the energy gradient. Only the non-negativity of $\alpha_{\mathcal{G}}$ is required for Lyapunov dissipation; $\beta_{\mathcal{G}}$ may be signed because the tangential term has zero instantaneous inner product with the energy gradient. We note that, while in general, there are many possible directions that are orthogonal to the gradient, in Section 3.2 we specify a procedure for learning this direction. In particular, we note that, by design, the first step decreases the energy, while the second is a tangential flow that preserves energy. Below, we formalize the tangential component and provide implementation details in Section 3.2.

Tangential Flow. Setting $\beta_{\mathcal{G}} = 0$ in Equation (2) yields a standard energy gradient flow applied to the features. While it guarantees energy dissipation, it may suffer from slow convergence (Boyd & Vandenberghe, 2004; Nocedal & Wright, 1999) and restricted dynamics during training. Pure gradient-flow samplers in generative settings can use hundreds or thousands of steps, which is less suitable for downstream graph learning because it corresponds to an extremely deep effective network, which is hard to train (Peng et al., 2024) and computationally expensive. To address this, and to accelerate the minimization of the energy function, we introduce a *tangential* flow that evolves tangentially to the gradient of $V_{\mathcal{G}}$, preserving energy. As we illustrate in Figure 1, and later theoretically discuss in Section 4, while the tangential flow itself maintains the same energy level, its combination with the energy gradient descent step, as shown in Equation (2), can offer a better overall descent direction, thereby accelerating energy convergence.

In order to obtain a direction that is orthogonal to $\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}(t))$, let $\mathbf{M}(\mathbf{H}(t))$ be a predicted update direction of the node features. We then define the *tangential* node feature update direction as:

$$T_{V_{\mathcal{G}}}(\mathbf{H}) = \mathbf{M}(\mathbf{H}) - \langle \mathbf{M}(\mathbf{H}), \mathbf{G}(\mathbf{H}) \rangle_F \mathbf{G}(\mathbf{H}), \quad (3)$$

where $\mathbf{G}(\mathbf{H}) = \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}) / \|\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H})\|_F$, $\langle \cdot, \cdot \rangle_F$ and $\|\cdot\|_F$ denote the Frobenius inner product and norm over the node-feature matrix, and we omit t for readability. If $\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}) = 0$, we define $T_{V_{\mathcal{G}}}(\mathbf{H}) = \mathbf{M}(\mathbf{H})$; in implementation, the denominator is stabilized with a small positive constant. When $\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}) \neq 0$, the projection in Equation (3) removes the component of $\mathbf{M}(\mathbf{H})$ in the energy-gradient direction and therefore enforces $\langle T_{V_{\mathcal{G}}}(\mathbf{H}), \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}) \rangle_F = 0$. Consequently, the tangential term cannot increase the instantaneous continuous-time energy derivative:

$$\begin{aligned} \frac{d}{dt} V_{\mathcal{G}}(\mathbf{H}(t)) &= \left\langle \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}(t)), \frac{d\mathbf{H}(t)}{dt} \right\rangle_F \\ &= -\alpha_{\mathcal{G}}(\mathbf{H}(t)) \|\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}(t))\|_F^2 \leq 0. \end{aligned} \quad (4)$$

3.2. TANGO Graph Neural Networks

In Section 3.1, we described the concept of TANGO and its underlying continuous dynamical system. To materialize this concept and obtain a GNN, we discretize Equation (2) using the commonly used in GNNs (Gravina et al., 2023; Eliasof et al., 2021; Chamberlain et al., 2021b; Arroyo et al., 2025; Choi et al., 2023) forward Euler approach to obtain the following graph neural layer:

$$\mathbf{H}^{(\ell+1)} = \mathbf{H}^{(\ell)} + \epsilon \left(-\alpha_{\mathcal{G}}(\mathbf{H}^{(\ell)}) \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}^{(\ell)}) + \beta_{\mathcal{G}}(\mathbf{H}^{(\ell)}) T_{V_{\mathcal{G}}}(\mathbf{H}^{(\ell)}) \right). \quad (5)$$

for $\ell = 0, \dots, L-1$, where $\epsilon > 0$ is a hyperparameter step size that stems from the forward Euler discretization that is commonly used in ODE-inspired GNNs (Chamberlain et al., 2021b; Eliasof et al., 2021; Gravina et al., 2023), further discussed in Appendix D.1, $\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}^{(\ell)})$ is the gradient of the energy function defined in Equation (8). The coefficients $\alpha_{\mathcal{G}} \geq 0$ and $\beta_{\mathcal{G}} \in \mathbb{R}$ are scalars that balance the energy descent and tangential terms, and are also predicted by the respective GNNs shown below. We learn these coefficients rather than treating them as fixed hyperparameters, to make the descent-tangential balance adaptive to the current graph, layer, and hidden state.

Energy Function. We now describe the implementation of $V_{\mathcal{G}}$. Given features $\mathbf{H}^{(\ell)}$, we apply:

$$\tilde{\mathbf{H}}^{(\ell)} = \sigma \left(\text{ENERGYGNN}(\mathbf{H}^{(\ell)}; \mathcal{G}) \right) \in \mathbb{R}^{n \times d}, \quad (6)$$

where ENERGYGNN is a graph neural network (e.g., GatedGCN (Bresson & Laurent, 2018), GPS (Rampášek et al., 2022)), and σ is a pointwise nonlinearity. We then compute per-node energy scores using a multilayer perceptron (MLP):

$$\tilde{V}_{\mathcal{G}}(\tilde{\mathbf{H}}^{(\ell)}) = \text{MLP}_{\mathcal{E}}(\tilde{\mathbf{H}}^{(\ell)}) \in \mathbb{R}^{n \times 1}, \quad (7)$$

and define the overall graph energy scalar value as:

$$V_{\mathcal{G}}(\mathbf{H}^{(\ell)}) = \frac{1}{n} \sum_{v \in \mathcal{V}} \tilde{V}_{\mathcal{G}}(\tilde{\mathbf{H}}^{(\ell)})_v^2 \in \mathbb{R}_{\geq 0}. \quad (8)$$

The parameters of ENERGYGNN and the energy MLP are updated through the downstream supervised loss (e.g., cross entropy or regression loss). Thus, $V_{\mathcal{G}}$ is a task-driven Lyapunov energy that shapes the hidden feature dynamics: the gradient term in Equation (5) provides energy descent, while the tangential term allows feature updates along its level sets. We also note that the gradient $\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}^{(\ell)})$ in Equation (5) is different from the gradient of the loss with respect to the parameters of ENERGYGNN and TANGENTGNN.

Thus, even if $\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}^{(\ell)})$ becomes small at some layer, the parameters still receive gradients through the dependence of $V_{\mathcal{G}}$ and $T_{V_{\mathcal{G}}}$ on their weights. Empirically, we do not observe vanishing gradients when increasing depth, as shown in Appendix E.2. Because $V_{\mathcal{G}}$ is optimized through the downstream objective, we view it as a latent task-driven potential that shapes hidden-state dynamics. In addition, we employ a global sum pooling (Xu et al., 2019) to $\tilde{\mathbf{H}}^{(\ell)}$, followed by an MLP and sigmoid activation, to obtain a bounded non-negative scalar $\alpha_{\mathcal{G}}$, as follows:

$$\alpha_{\mathcal{G}}(\mathbf{H}^{(\ell)}) = \text{SIGMOID} \left(\text{MLP}_{\alpha} \left(\text{SUMPOOL}(\tilde{\mathbf{H}}^{(\ell)}) \right) \right) \quad (9)$$

We note that non-negativity is required for a valid gradient descent to be obtained in Equation (5), and the bounded value is chosen to maintain stable training. Applying sigmoid to $\alpha_{\mathcal{G}}$ thus acts as a learned gate on the dissipative part of the dynamics: the model can strengthen descent when energy reduction is useful and attenuate it when the representation should mainly move along an energy level set.

Tangential Update. To compute the tangential update $T_{V_{\mathcal{G}}}(\mathbf{H}^{(\ell)})$, we learn a dedicated GNN denoted by TANGENTGNN. Specifically, given input features $\mathbf{H}^{(\ell)}$, we predict a node feature update:

$$\mathbf{M}^{(\ell)} = \sigma \left(\text{TANGENTGNN}(\mathbf{H}^{(\ell)}; \mathcal{G}) \right), \quad (10)$$

and define the energy-tangential component via orthogonal projection, as described in Equation (3). Also, we define the scalar $\beta_{\mathcal{G}}$ that scales the tangential term, as follows:

$$\beta_{\mathcal{G}}(\mathbf{H}^{(\ell)}) = \text{MLP}_{\beta} \left(\text{SUMPOOL}(\mathbf{M}^{(\ell)}) \right) \in \mathbb{R}. \quad (11)$$

Different than $\alpha_{\mathcal{G}}$, the coefficient $\beta_{\mathcal{G}}$ is not required to be non-negative for the continuous-time Lyapunov construction, because the tangential term is orthogonal to the energy gradient. Its magnitude and sign determine how the representation moves along the learned level sets.

Backbone choice. The Lyapunov and tangential-flow construction is architecture-agnostic: ENERGYGNN and TANGENTGNN can be instantiated with different message-passing or transformer-style graph backbones. The theoretical properties rely on the scalar energy, the non-negative descent gate, and the orthogonal projection, while practical performance also inherits the optimization behavior and inductive biases of the chosen backbone. In our experiments, we evaluate TANGO paired with GCN, GIN, GatedGCN, and GPS.

4. Theoretical Properties of TANGO

We now analyze the continuous-time dynamics of TANGO as defined in Equation (2). Our analysis focuses on three

aspects: *energy dissipation, feature evolution in flat energy landscapes, and the benefit of the tangent direction*. Proofs are provided in Appendix B.

Assumptions and Notations. Throughout this analysis, we assume that: (i) the input graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is connected; (ii) the energy function $V_{\mathcal{G}}(\mathbf{H}(t))$ is twice differentiable and bounded from below. For simplicity of notation, throughout this section we omit the time or layer scripts, and use the term $\mathbf{H}$ to denote node features, when possible.

We start by showing that TANGO is dissipative if $\|\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H})\|^2 > 0$, and $\alpha_{\mathcal{G}} \geq 0$ (obtained by design), corresponding to the Lyapunov stability criterion from Theorem 2.2.

Proposition 4.1 (Energy is non-increasing). *Suppose $\alpha_{\mathcal{G}} \geq 0$ and $\|\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H})\|^2 > 0$. Then the energy $V_{\mathcal{G}}(\mathbf{H})$ is non-increasing along trajectories of Equation (2). Specifically,*

$$\begin{aligned} \frac{d}{dt} V_{\mathcal{G}}(\mathbf{H}) &= -\alpha_{\mathcal{G}}(\mathbf{H}) \|\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H})\|^2 \\ &\quad + \beta_{\mathcal{G}}(\mathbf{H}) \langle T_{V_{\mathcal{G}}}(\mathbf{H}), \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}) \rangle \\ &= -\alpha_{\mathcal{G}}(\mathbf{H}) \|\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H})\|^2 \leq 0. \end{aligned} \quad (12)$$

Proposition 4.1 establishes a standard Lyapunov property: the energy $V_{\mathcal{G}}(\mathbf{H})$ is non-increasing along trajectories (i.e., layers), and, by construction, is bounded from below. As a result, trajectories remain in sublevel sets $\{\mathbf{H} : V_{\mathcal{G}}(\mathbf{H}) \leq V_{\mathcal{G}}(\mathbf{H}^{(0)})\}$. Notably, this energy-control result shows that the tangential component $T_{V_{\mathcal{G}}}(\mathbf{H})$ can still move features along level sets of $V_{\mathcal{G}}$, including in regions where the energy gradient is small. We now show that unlike gradient flows, our TANGO admits evolution of node features in flat energy landscapes, a prime challenge in optimization techniques (Nocedal & Wright, 1999; Boyd & Vandenberghe, 2004).

Proposition 4.2 (TANGO can Evolve Features in Flat Energy Landscapes). *Suppose $\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}) = 0$, and $T_{V_{\mathcal{G}}}(\mathbf{H}) \neq 0$, then the TANGO flow in Equation (2) reads:*

$$\frac{d\mathbf{H}}{dt} = \beta_{\mathcal{G}}(\mathbf{H}) T_{V_{\mathcal{G}}}(\mathbf{H}).$$

This implies that in contrast to gradient flows, the dynamics of TANGO obtained by the tangential term can evolve even in regions where the energy landscape is flat.

Theoretical Benefits of Using the Tangent Direction. Our TANGO combines two terms as shown in Equation (2) and its discretization in Equation (5). These are the energy gradient $\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}^{(\ell)})$ and the tangential direction vector $T_{V_{\mathcal{G}}}(\mathbf{H})$. A natural theoretical and practical question is: *under what conditions does the inclusion of the tangential direction improve over simple gradient descent?* To address this question, we first recall a classic convergence result for gradient-based minimization.

Proposition 4.3 (Convergence of Gradient Descent of a Scalar Function, Nocedal & Wright (1999)). *Let $V_{\mathcal{G}}(\cdot)$ be a scalar function and let $\mathbf{H}^{(\ell+1)} = \mathbf{H}^{(\ell)} - \alpha_{\mathcal{G}}^{(\ell)}(\mathbf{H}^{(\ell)}) \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}^{(\ell)})$ be a gradient-descent iteration of the energy $V_{\mathcal{G}}(\cdot)$. Then, a linear convergence is obtained, with a convergence rate:*

$$r = \frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}},$$

where $\lambda_{\max}$ is the maximal eigenvalue, and in the case of problems that involve the graph Laplacian, $\lambda_{\min}$ is the second minimal eigenvalue, i.e., the first non-zero eigenvalue of the Hessian of $V_{\mathcal{G}}(\cdot)$.

Proposition 4.3 shows that gradient descent suffers in ill-conditioned problems, i.e., when the ratio between the $\lambda_{\max}$ and $\lambda_{\min}$ is large. This is common in graph-based tasks, where the Hessian may inherit poor conditioning from the graph Laplacian, particularly when oversquashing occurs due to bottlenecks in the graph (Topping et al., 2022; Giraldo et al., 2023; Di Giovanni et al., 2023). Alternatively, consider the effect of adding an orthogonal flow to the gradient descent direction. Then, the update direction is

$$\mathbf{D} = \alpha_{\mathcal{G}}(\mathbf{H}^{(\ell)}) \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}^{(\ell)}) + \beta_{\mathcal{G}}(\mathbf{H}^{(\ell)}) T_{V_{\mathcal{G}}}(\mathbf{H}^{(\ell)}). \quad (13)$$

The following proposition demonstrates that it is possible, i.e., the model has the capacity to learn T such that $\mathbf{D}$ becomes a Newton-like direction with quadratic convergence (Nocedal & Wright, 1999).

Proposition 4.4 (TANGO can learn a Quadratic Convergence Direction). *Assume for simplicity that $\beta_{\mathcal{G}} = 1$, and that the Hessian of $V_{\mathcal{G}}$ is invertible. Let $\mathbf{D} = \alpha_{\mathcal{G}}(\mathbf{H}^{(\ell)}) \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}^{(\ell)}) + T_{V_{\mathcal{G}}}(\mathbf{H}^{(\ell)})$ with $\langle T_{V_{\mathcal{G}}}(\mathbf{H}^{(\ell)}), \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}^{(\ell)}) \rangle = 0$. Then, it is possible to learn a direction $T_{V_{\mathcal{G}}}(\mathbf{H}^{(\ell)})$ and a step size $\alpha_{\mathcal{G}}$ such that $\mathbf{D}$ is the Newton direction, $\mathbf{N} = (\nabla^2 V_{\mathcal{G}})^{-1} \nabla V_{\mathcal{G}}$.*

In addition to its improved global convergence, Newton’s method is notable for its local convergence rate behavior, being independent of the condition number of the Hessian (Nocedal & Wright, 1999; Boyd & Vandenberghe, 2004). This implies that if the tangential flow is learned to approximate Newton direction, TANGO can overcome the slow convergence caused by highly ill-conditioned energy landscapes, as commonly observed in different second-order optimization techniques and their approximations, such as conjugate gradients (CG) and LBFGS (Nocedal & Wright, 1999; Boyd & Vandenberghe, 2004). In the context of graph learning, Proposition 4.4 is particularly relevant when considering the oversquashing problem (Alon & Yahav, 2021; Di Giovanni et al., 2023), and motivates the utilization of the conceptual blueprint of TANGO from Equation (2) for graph learning: oversquashing leads to poor conditioning;

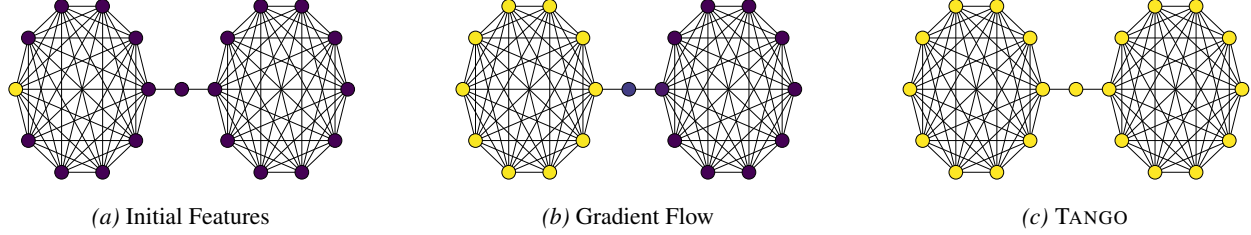


Figure 2. Comparison of propagation behaviors between gradient flow and TANGO with 50 layers. While a Dirichlet-energy gradient flow struggles to propagate information through the bottleneck, TANGO propagates the signal more effectively. Additional details and diagnostics are provided in Appendix D.2.

the graph Laplacian has a smallest eigenvalue of zero (for connected graphs), and the second smallest eigenvalue is also close to zero (Topping et al., 2022; Giraldo et al., 2023; Black et al., 2023; Jamadandi et al., 2024). Under these conditions, gradient flow methods, which are implicitly implemented by common GNN formulations (Giovanni et al., 2023), perform poorly due to their ill-conditioned energy landscape, limiting the ability to propagate information between nodes. By enabling feature updates that can approximate second-order information, i.e., Newton-like directions, our TANGO offers a mechanism that can, in principle, alleviate oversquashing. We emphasize that this is a mechanism and capacity statement, not a guarantee that every learned instance eliminates oversquashing on every graph. The role of the theory is to identify the regime in which TANGO should be useful: Proposition 4.1 gives Lyapunov energy control, the flat-region result shows that tangential motion can continue when pure descent stalls, the gradient-descent convergence bound identifies ill-conditioning as a failure mode for first-order dynamics, and Proposition 4.4 shows that the combined direction can represent improved update directions. Graph bottlenecks instantiate this poorly conditioned communication regime, and the diagnostics in Tables 6 and 7 provide direct empirical evidence for this mechanism. We empirically validate these results in Figure 2, where we compare TANGO with a Dirichlet energy gradient flow process, which is often implemented by baseline GNNs (Rusch et al., 2023; Giovanni et al., 2023). Additional bottleneck and one-step diagnostics are provided in Appendix D.2, and broader benchmark results are reported in Section 5.

5. Experiments

We evaluate the performance of our TANGO on a suite of benchmarks: (i) synthetic benchmarks that require the exchange of messages with large distances, called graph property prediction from Gravina et al. (2023), in Section 5.1; (ii) the peptides long-range graph benchmark (Dwivedi et al., 2022b) in Section 5.2; (iii) GNN benchmarks from (Dwivedi et al., 2023) including the ZINC-12k, MNIST,

Table 1. Mean test set $\log_{10}(\text{MSE})$ ($\downarrow$) and std on 4 random weight initializations on Graph Property Prediction. Lower is better. **First**, **second**, and **third** best results for each task are color-coded.

Model	Diameter	SSSP	Eccentricity
MPNNs			
GatedGCN	0.1348 ± 0.0397	-3.2610 ± 0.0514	0.6995 ± 0.0302
GCN	0.7424 ± 0.0466	0.9499 ± 0.0001	0.8468 ± 0.0028
GAT	0.8221 ± 0.0752	0.6951 ± 0.1499	0.7909 ± 0.0222
GraphSAGE	0.8645 ± 0.0401	0.2863 ± 0.1843	0.7863 ± 0.0207
GIN	0.6131 ± 0.0990	-0.5408 ± 0.4193	0.9504 ± 0.0007
GCNII	0.5287 ± 0.0570	-1.1329 ± 0.0135	0.7640 ± 0.0355
DE-GNNs			
DGC	0.6028 ± 0.0050	-0.1483 ± 0.0231	0.8261 ± 0.0032
GRAND	0.6715 ± 0.0490	-0.0942 ± 0.3897	0.6602 ± 0.1393
GraphCON	0.0964 ± 0.0620	-1.3836 ± 0.0092	0.6833 ± 0.0074
A-DGN	-0.5188 ± 0.1812	-3.2417 ± 0.0751	0.4296 ± 0.1003
SWAN	-0.5981 ± 0.1145	-3.5425 ± 0.0830	-0.0739 ± 0.2190
PH-DGN	-0.5385 ± 0.0187	-4.2993 ± 0.0721	-0.9348 ± 0.2097
Transformers			
GPS	-0.5121 ± 0.0426	-3.5990 ± 0.1949	0.6077 ± 0.0282
Ours			
TANGO _{GCN}	0.1729 ± 0.0382	-1.0024 ± 0.0854	-1.6264 ± 0.0053
TANGO _{GIN}	0.0433 ± 0.0211	-2.8923 ± 0.0937	-1.7228 ± 0.0046
TANGO _{GATEDGCN}	-0.6681 ± 0.0745	-5.0626 ± 0.0742	-1.7419 ± 0.0106
TANGO _{GPS}	-0.9772 ± 0.0518	-5.5263 ± 0.0838	-2.1455 ± 0.0033

CIFAR-10, PATTERN, and CLUSTER datasets; and (iv) the heterophilic node classification datasets from Platonov et al. (2023). Notably, TANGO shows consistent downstream performance improvements over its four backbone models: GCN (Kipf & Welling, 2017), GIN (Xu et al., 2019), GatedGCN (Bresson & Laurent, 2018), and GPS (Rampáček et al., 2022), highlighting its practicality. It also offers competitive performance compared with other popular and state-of-the-art methods, such as MPNN-based models, DE-GNNs, higher-order DGNs, and graph transformers. In all experiments, TANGO is trained with the same loss function as other GNN baselines, like the cross-entropy loss. In Appendix D we provide full experimental details on the hyperparameters, benchmark evaluation, and runtimes. Additional results and comparisons, as well as evaluations of heterophilic node classification and ablation studies that iso-

Table 2. Test performance in five benchmarks from (Dwivedi et al., 2023). Shown is the mean $\pm$ std of 4 runs with different random seeds. Highlighted are the top **first**, **second**, and **third** results.

Model	ZINC-12k	MNIST	CIFAR10	PATTERN	CLUSTER
	MAE↓	Accuracy↑	Accuracy↑	Accuracy↑	Accuracy↑
GCN (Kipf & Welling, 2017)	0.367 $\pm$ 0.011	90.705 $\pm$ 0.218	55.710 $\pm$ 0.381	71.892 $\pm$ 0.334	68.498 $\pm$ 0.976
GIN (Xu et al., 2019)	0.526 $\pm$ 0.051	96.485 $\pm$ 0.252	55.255 $\pm$ 1.527	85.387 $\pm$ 0.136	64.716 $\pm$ 1.553
GAT (Veličković et al., 2018)	0.384 $\pm$ 0.007	95.535 $\pm$ 0.205	64.223 $\pm$ 0.455	78.271 $\pm$ 0.186	70.587 $\pm$ 0.447
GatedGCN (Bresson & Laurent, 2018)	0.282 $\pm$ 0.015	97.340 $\pm$ 0.143	67.312 $\pm$ 0.311	85.568 $\pm$ 0.088	73.840 $\pm$ 0.326
PNA (Corso et al., 2020)	0.188 $\pm$ 0.004	97.940 $\pm$ 0.120	70.350 $\pm$ 0.630	—	—
DGN (Beaini et al., 2021)	0.168 $\pm$ 0.003	—	72.838 $\pm$ 0.417	86.680 $\pm$ 0.034	—
CRaW1 (Tönshoff et al., 2023b)	0.085 $\pm$ 0.004	97.944 $\pm$ 0.050	69.013 $\pm$ 0.259	—	—
GIN-AK+ (Zhao et al., 2022)	0.080 $\pm$ 0.001	—	72.190 $\pm$ 0.130	86.850 $\pm$ 0.057	—
SAN (Kreuzer et al., 2021)	0.139 $\pm$ 0.006	—	—	86.581 $\pm$ 0.037	76.691 $\pm$ 0.65
EGT (Hussain et al., 2022)	0.108 $\pm$ 0.009	98.173 $\pm$ 0.087	68.702 $\pm$ 0.409	86.821 $\pm$ 0.020	79.232 $\pm$ 0.348
Graphormer-GD (Zhang et al., 2023)	0.081 $\pm$ 0.009	—	—	—	—
GPS (Rampášek et al., 2022)	0.070 $\pm$ 0.004	98.051 $\pm$ 0.126	72.298 $\pm$ 0.356	86.685 $\pm$ 0.059	78.016 $\pm$ 0.180
GRIT (Ma et al., 2023)	0.059 $\pm$ 0.002	98.108 $\pm$ 0.111	76.468 $\pm$ 0.881	87.196 $\pm$ 0.076	80.026 $\pm$ 0.277
TANGO _{GCN}	0.153 $\pm$ 0.010	94.579 $\pm$ 0.211	64.920 $\pm$ 0.402	81.198 $\pm$ 0.299	74.040 $\pm$ 1.109
TANGO _{GIN}	0.122 $\pm$ 0.031	97.651 $\pm$ 0.247	66.350 $\pm$ 0.967	86.703 $\pm$ 0.194	71.360 $\pm$ 1.169
TANGO _{GatedGCN}	0.128 $\pm$ 0.011	97.788 $\pm$ 0.105	70.894 $\pm$ 0.329	86.672 $\pm$ 0.071	78.194 $\pm$ 0.307
TANGO _{GPS}	0.062 $\pm$ 0.005	98.197 $\pm$ 0.110	75.783 $\pm$ 0.261	87.182 $\pm$ 0.063	80.113 $\pm$ 0.138

late the effects of the energy term, the tangential projection, and depth, are provided in Appendix E.

5.1. Graph Property Prediction

Setup. We consider the three graph property prediction tasks from Gravina et al. (2023), evaluating the performance of TANGO in predicting graph diameters, single source shortest paths (SSSP), and node eccentricity on synthetic graphs. To effectively address these tasks, it is essential to propagate information not only from direct neighbors but also from distant nodes within the graph. As a result, strong performance in these tasks mirrors the ability to facilitate long-range interactions.

Results. Table 1 reports the mean test $\log_{10}(\text{MSE})$, comparing our TANGO with various MPNNs, DE-GNNs, and transformer-based models. The results highlight that TANGO, in all variants, consistently achieves the lowest (best) error across all tasks, demonstrating its efficacy compared with existing methods. For example, in the Eccentricity task, TANGO_{GPS} reduces the error score by over 1.2 points compared to PH-DGN (Heilig et al., 2025) and by over 2.0 points compared to SWAN, which are models designed to propagate information over long radii effectively. On Diameter and SSSP, TANGO_{GPS} also yields gains over the strong prior DE-GNN baseline PH-DGN, improving the $\log_{10}(\text{MSE})$ by 0.4 and 1.2 points respectively. Overall, these results validate the effectiveness of our TANGO in modeling long-range interactions and are consistent with alleviation of oversquashing. Furthermore, TANGO strengthens the performance of simple MPNN backbones like GatedGCN.

For example, GatedGCN augmented with our TANGO consistently delivers better results than the baseline GatedGCN, highlighting its ability to enhance traditional MPNNs. This demonstrates that our method can effectively leverage the strengths of simple models while overcoming their limitations in long-range propagation.

5.2. Long-Range Benchmark

Setup. We evaluate our method on the real-world Long-Range Graph Benchmark (LRGB) (Dwivedi et al., 2022b), focusing on *Peptides-func* and *Peptides-struct*. We follow the experimental setting in (Dwivedi et al., 2022b), including the 500K parameter budget. Transformer baselines use positional and structural encodings; TANGO uses none. The datasets contain large peptide molecular graphs, whose structure and function depend on long-range interactions. Thus, short-range interactions, such as local message passing in GNNs, may be insufficient for this task.

Results. Table 3 compares our TANGO model with a wide range of baselines. For example, on *Peptides-struct*, all TANGO variants achieve competitive MAE in Table 3 under the shared parameter budget. A broader comparison is presented in Table 14. The results indicate that TANGO outperforms or offers competitive performance compared with standard MPNNs, transformer-based GNNs, DE-GNNs, Multi-hop GNNs, and methods that use rewiring (Chamberlain et al., 2021b; Gutteridge et al., 2023).

Table 3. Results for Peptides-func and Peptides-struct (3 training seeds). The **first**, **second**, and **third** best scores are colored.

Model	Peptides-func AP $\uparrow$	Peptides-struct MAE $\downarrow$
MPNNs		
GCN	59.30 $\pm$ 0.23	0.3496 $\pm$ 0.0013
GINE	54.98 $\pm$ 0.79	0.3547 $\pm$ 0.0045
GCNII	55.43 $\pm$ 0.78	0.3471 $\pm$ 0.0010
GatedGCN	58.64 $\pm$ 0.77	0.3420 $\pm$ 0.0013
Multi-hop GNNs		
DIGL+MPNN+LapPE	68.30 $\pm$ 0.26	0.2616 $\pm$ 0.0018
MixHop-GCN+LapPE	68.43 $\pm$ 0.49	0.2614 $\pm$ 0.0023
DRew-GCN+LapPE	71.50$\pm$0.44	0.2536 $\pm$ 0.0015
Transformers		
Transformer+LapPE	63.26 $\pm$ 1.26	0.2529 $\pm$ 0.0016
SAN+LapPE	63.84 $\pm$ 1.21	0.2683 $\pm$ 0.0043
GPS+LapPE	65.35 $\pm$ 0.41	0.2500 $\pm$ 0.0005
DE-GNNs		
GRAND	57.89 $\pm$ 0.62	0.3418 $\pm$ 0.0015
GraphCON	60.22 $\pm$ 0.68	0.2778 $\pm$ 0.0018
A-DGN	59.75 $\pm$ 0.44	0.2874 $\pm$ 0.0021
SWAN	67.51 $\pm$ 0.39	0.2485$\pm$0.0009
PH-DGN	70.12$\pm$0.45	0.2465$\pm$0.0020
Ours		
TANGO _{GCN}	69.17 $\pm$ 0.31	0.2432 $\pm$ 0.0011
TANGO _{GIN}	68.78 $\pm$ 0.66	0.2440 $\pm$ 0.0024
TANGO _{GATEDGCN}	68.92 $\pm$ 0.40	0.2451 $\pm$ 0.0006
TANGO _{GPS}	70.21$\pm$0.43	0.2422$\pm$0.0014

5.3. Heterophilic Node Classification

Setup. We consider heterophilic node classification datasets; *Roman-empire*, *Amazon-ratings*, *Minesweeper*, *Tolokers*, and *Questions* tasks, to evaluate TANGO in capturing complex node relationships beyond homophily. We use the training and evaluation protocols from Platonov et al. (2023).

Results. We report the performance of TANGO in Appendix E.1, and compare it with several recent leading methods. Specifically, we include baseline results from (Finkelshtein et al., 2024; Platonov et al., 2023; Müller et al., 2024). Across all datasets, TANGO achieves competitive performance that often outperforms state-of-the-art methods, and consistently improves its backbone GNN performance, demonstrating that our TANGO can also be utilized on larger graphs and in heterophilic scenarios.

6. Related Work

We now cover two main topics related to our TANGO, with additional related works in Appendix A.

Deep GNNs and Dynamical Systems. A growing body of work interprets GNN layers as iterative updates in a dynamical system, providing a principled framework for analyzing stability, controlling diffusion, and informing architectural

design. Poli et al. (2019) introduced Graph Neural ODEs, inspired by neural ODEs (Ruthotto & Haber, 2020; Chen et al., 2018), modeling node feature evolution via continuous-depth ODEs aligned with graph structure, enabling adaptive computation and improved performance in dynamic settings. Similarly, Xhonneux et al. (2020) proposed Continuous GNNs, where feature channels evolve by differential equations, mitigating over-smoothing via infinite-depth limits. Follow-up works such as GODE (Zhuang et al., 2020), GRAND (Chamberlain et al., 2021b), PDE-GCN_D (Eliasof et al., 2021), and DGC (Wang et al., 2021) view GNN layers as discrete integration steps of the heat equation to control oversmoothing (Nt & Maehara, 2019; Oono & Suzuki, 2020; Cai & Wang, 2020). Extensions like PDE-GCN_M (Eliasof et al., 2021) and GraphCON (Rusch et al., 2022) add oscillatory components to preserve feature energy, while others leverage heat-kernel attention (Choromanski et al., 2022), anti-symmetry (Gravina et al., 2023; 2025), reaction-diffusion (Wang et al., 2023; Choi et al., 2023), advection-reaction-diffusion (Eliasof et al., 2024a) to enhance long-range or directional flow, and higher-order graph neuro ODE models (Eliasof et al., 2024b). A comprehensive overview is given in Han et al. (2023). Closely related, Giovanni et al. (2023) interpret GNN layer updates as gradient flows of the Dirichlet energy, aligning message passing with energy minimization, and Zhao et al. (2023) studies the adversarial robustness of Lyapunov-based GNNs. In contrast, our TANGO takes a different approach. Instead of viewing GNN layers as discretizations whose weights are to be learned, TANGO learns a graph-adaptive, task-driven energy and introduces a novel descent mechanism combining energy gradients with a learnable tangential component, enabling more flexible dynamics than pure gradient flows.

Learning Energy Functions in Neural Networks. Energy-based models (EBMs) provide a flexible framework in deep learning by learning an energy function whose low-energy regions correspond to regions of high probability in the data. They have been widely used in generative tasks such as image synthesis (LeCun et al., 2006; Xie et al., 2016; Du & Mordatch, 2019; Guo et al., 2023) and graph generation (Liu et al., 2021). In contrast to these typically unsupervised settings, our work uses a *task-driven* energy function whose parameters are optimized only through supervised losses on node or graph labels, rather than via a separate generative objective. The energy serves as a Lyapunov potential that shapes the dynamics of hidden features; we do not require or assume that its global minima coincide with globally optimal predictions. Relatedly, Lyapunov functions, classical tools from control theory (Khalil, 2002), have been used in neural networks to ensure stable learning or inference dynamics, e.g., by enforcing stability in Neural ODEs (Rodriguez et al., 2022) and GNN-based controllers (Fallin et al., 2025), Hamiltonian graph flows for adversarial robust-

ness (Zhao et al., 2023), as well as Beltrami flow and neural diffusion on graphs (Chamberlain et al., 2021a), which use a discretization of the Beltrami flow in joint feature and position space and induce an implicit rewiring mechanism. Notably, in a Hamiltonian system, the energy is conserved, which has been shown to be useful for adversarial robustness (Zhao et al., 2023), while a Lyapunov-stable system implies that close initial conditions evolve along similar trajectories. Lyapunov-stable neural ODEs have also been studied by regularizing an ODE to obtain Lyapunov-stable equilibria, an approach that has been found beneficial for adversarial robustness in image classification (Kang et al., 2021). Our TANGO is complementary to these lines of work: it operates directly on graph-structured hidden states, learns a task-driven graph energy that is used explicitly as a Lyapunov function, and couples its gradient flow with a learned tangential component. This tangential flow can both accelerate energy minimization and maintain informative feature updates even in areas where the energy landscape is flat. In this way, TANGO bridges and extends these perspectives by introducing a graph-adaptive, task-specific energy and a novel feature evolution mechanism, reflected in its enhanced downstream performance on graph learning tasks, as shown in Section 5.

7. Conclusions

We introduced TANGO, a novel framework for learning graph neural dynamics by jointly modeling an energy descent direction and a tangential flow. By interpreting GNN message passing through the lens of Lyapunov theory and continuous dynamical systems, TANGO unifies task-driven energy-based modeling with flexible, learnable tangential flows, which allow for better utilization of the learned energy function by accelerating its minimization. We further show that the tangential component enables continued feature evolution in flat or ill-conditioned energy landscapes, offering a compelling advantage over traditional gradient flow approaches. We relate this property to the mitigation of oversquashing, a persistent challenge in graph learning. Empirically, TANGO achieves strong performance across 15 synthetic and real-world benchmarks, outperforming message-passing, diffusion-based, and attention-based GNNs. This work opens several interesting directions for future research, including the incorporation of higher-order differential operators into the tangential flow mechanism, the development of analysis and regularization techniques for the learned energy landscape, and the study of tangential flows in other domains and applications.

Limitations. While our results show that TANGO improves long-range propagation across several bottleneck-sensitive benchmarks accompanied by theoretical understanding of the added capacity via the tangential term, oversquashing

alleviation is supported empirically. The learned energy is used to shape hidden-state dynamics, and a post-hoc interpretation of the energy landscape is an interesting direction for future work. Finally, TANGO adds the cost of learning both energy and tangential update networks and, like other backbone-based graph learning frameworks, its performance can depend on the chosen backbone, depth, and discretization choices.

Impact Statement

This paper presents work whose goal is to advance the field of Graph Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Additional Related Work

Oversquashing in Graph Learning. Graph neural networks (GNNs) typically operate via message-passing mechanisms that aggregate information from local neighborhoods. While effective in capturing short-range dependencies, this design often leads to *oversquashing*, a phenomenon where signals from distant nodes are compressed into fixed-size representations, impeding the flow of long-range information (Alon & Yahav, 2021; Di Giovanni et al., 2023; Topping et al., 2022). This limitation poses a challenge in domains that demand rich global context, such as bioinformatics (Baek et al., 2021; Dwivedi et al., 2022b) and heterophilic graphs (Luan et al., 2024; Wang et al., 2024b). A range of strategies has been proposed to mitigate oversquashing. *Graph rewiring* approaches, such as SDRF (Topping et al., 2022), densify the graph to enhance connectivity prior to training. In contrast, methods like GRAND (Chamberlain et al., 2021b), BLEND (Chamberlain et al., 2021a), and DRew (Gutteridge et al., 2023) adjust the graph structure dynamically based on node features. *Transformer-based models* offer another promising route by leveraging global attention to enable direct, long-range message passing. Examples include SAN (Kreuzer et al., 2021), Graphormer (Ying et al., 2021), and GPS (Rampášek et al., 2022), which incorporate positional encodings, such as Laplacian eigenvectors (Dwivedi et al., 2023) and random walk structural embeddings (Dwivedi et al., 2022a) to preserve structural identity. However, the quadratic complexity of full attention in these models raises scalability concerns, motivating interest in sparse attention mechanisms (Zaheer et al., 2020; Choromanski et al., 2021; Shirzad et al., 2023). An alternative line of work explores *non-local dynamics* to enhance expressivity without relying solely on attention. FLODE (Maskey et al., 2023) employs fractional graph operators, QDC (Markovich, 2023) uses quantum diffusion processes, and G2TN (Toth et al., 2022) models explicit diffusion paths to propagate information more effectively. While these approaches address the oversquashing bottleneck, they often come with increased computational demands due to dense propagation operators. For a broader overview of these techniques, see Shi et al. (2023). We note that the challenge of modeling long-range dependencies also arises in other domains, such as sequential architectures (Hochreiter & Schmidhuber, 1997; Gu et al., 2022).

Optimization Techniques. The formulation of TANGO parallels concepts explored in the optimization literature, particularly in the design of dynamical systems that balance expressivity and convergence. While traditional gradient descent provides a robust and interpretable mechanism for minimizing energy functions, its convergence rate can be limited in poorly conditioned settings (Boyd & Vandenberghe, 2004; Nocedal & Wright, 1999), which frequently arise in graph-based problems due to structural bottlenecks (Alon & Yahav, 2021; Topping et al., 2022). Second-order approaches, such as Newton’s method, are known to accelerate convergence by incorporating curvature information, albeit at increased computational cost. The combination of energy gradient descent and a learned tangential component in TANGO suggests a learnable departure from purely first-order schemes. Rather than explicitly computing or approximating the Hessian, our framework enables the model to learn corrective update directions that are orthogonal to the descent path. This design implicitly aligns with the motivations behind quasi-Newton techniques like conjugate gradients and LBFGS (Nocedal & Wright, 1999), which aim to improve convergence by leveraging directional information that complements the gradient. From this perspective, TANGO can be viewed as embedding optimization-inspired dynamics within graph learning frameworks. This is particularly relevant in scenarios affected by oversquashing (Di Giovanni et al., 2023), where effective feature transmission often requires departing from strictly local, gradient-driven updates. By allowing energy-preserving tangential flows, TANGO introduces flexibility reminiscent of structured optimization methods, adapted to the graph learning domain.

B. Proofs of Theoretical Results

In this section, we restate the theoretical results from Section 4 and provide their proofs. As in the main text, we assume the following throughout: (i) the input graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is connected; (ii) the energy function $V_{\mathcal{G}}(\mathbf{H}(t))$ is twice differentiable and bounded from below. For simplicity of notation, throughout this section, we omit the time or layer scripts and use the term $\mathbf{H}$ to denote node features when possible.

Proposition B.1 (Energy Dissipation). *Suppose $\alpha_{\mathcal{G}} \geq 0$ and $\|\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H})\|^2 > 0$. Then the energy $V_{\mathcal{G}}(\mathbf{H})$ is non-increasing along trajectories of Equation (2). Specifically,*

$$\begin{aligned} \frac{d}{dt} V_{\mathcal{G}}(\mathbf{H}) &= -\alpha_{\mathcal{G}}(\mathbf{H}) \|\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H})\|^2 + \beta_{\mathcal{G}}(\mathbf{H}) \langle T_{V_{\mathcal{G}}}(\mathbf{H}), \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}) \rangle \\ &= -\alpha_{\mathcal{G}}(\mathbf{H}) \|\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H})\|^2 \leq 0. \end{aligned}$$

Proof. By the chain rule,

$$\frac{d}{dt} V_{\mathcal{G}}(\mathbf{H}) = \left\langle \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}), \frac{d\mathbf{H}}{dt} \right\rangle.$$

Substituting the dynamics of Equation (2):

$$\begin{aligned} \frac{d}{dt} V_{\mathcal{G}}(\mathbf{H}) &= \langle \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}), -\alpha_{\mathcal{G}}(\mathbf{H}) \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}) + \beta_{\mathcal{G}}(\mathbf{H}) T_{V_{\mathcal{G}}}(\mathbf{H}) \rangle \\ &= -\alpha_{\mathcal{G}}(\mathbf{H}) \|\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H})\|^2 + \beta_{\mathcal{G}}(\mathbf{H}) \langle T_{V_{\mathcal{G}}}(\mathbf{H}), \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}) \rangle. \end{aligned}$$

As discussed in Section 3, we have by design, that

$$\langle T_{V_{\mathcal{G}}}(\mathbf{H}), \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}) \rangle = 0.$$

Therefore,

$$\frac{d}{dt} V_{\mathcal{G}}(\mathbf{H}) = -\alpha_{\mathcal{G}}(\mathbf{H}) \|\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H})\|^2.$$

Because $\alpha_{\mathcal{G}}(\mathbf{H}) \geq 0$ by design, the energy is non-increasing, and assuming $\alpha_{\mathcal{G}}(\mathbf{H}) > 0$, the system is dissipative, i.e., its energy is decreasing. $\square$

Proposition B.2 (TANGO can Evolve Features in Flat Energy Landscapes). *Suppose $\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}) = 0$, and $T_{V_{\mathcal{G}}}(\mathbf{H}) \neq 0$, then the TANGO flow in Equation (2) reads:*

$$\frac{d\mathbf{H}}{dt} = \beta_{\mathcal{G}}(\mathbf{H}) T_{V_{\mathcal{G}}}(\mathbf{H}).$$

This implies that in contrast to gradient flows, the dynamics of TANGO can evolve even in regions where the energy landscape is flat.

Proof. Because $\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}) = 0$, the first term in Equation (2) vanishes, and the TANGO dynamical system reads:

$$\frac{d\mathbf{H}}{dt} = \beta_{\mathcal{G}}(\mathbf{H}) T_{V_{\mathcal{G}}}(\mathbf{H}),$$

Assuming that $T_{V_{\mathcal{G}}}(\mathbf{H}) \neq 0$, TANGO can continue evolving node features also in cases where $\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}) = 0$, i.e., where the energy landscape is flat. $\square$

Proposition B.3 (Convergence of Gradient Descent of a Scalar Function, Nocedal & Wright (1999)). *Let $V_{\mathcal{G}}(\cdot)$ be a scalar function and let $\mathbf{H}^{(\ell+1)} = \mathbf{H}^{(\ell)} - \alpha_{\mathcal{G}}^{(\ell)}(\mathbf{H}^{(\ell)}) \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}^{(\ell)})$ be a gradient-descent iteration of the energy $V_{\mathcal{G}}(\cdot)$. Then, a linear convergence is obtained, with a convergence rate:*

$$r = \frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}},$$

where $\lambda_{\max}$ is the maximal eigenvalue, and in the case of problems that involve the graph Laplacian, $\lambda_{\min}$ is the second minimal eigenvalue, i.e., the first non-zero eigenvalue of the Hessian of $V_{\mathcal{G}}(\cdot)$.

Proposition B.4 (TANGO can learn a Quadratic Convergence Direction). *Assume for simplicity that $\beta_{\mathcal{G}} = 1$, and that the Hessian of $V_{\mathcal{G}}$ is invertible. Let $\mathbf{D} = \alpha_{\mathcal{G}}(\mathbf{H}^{(\ell)}) \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}^{(\ell)}) + T_{V_{\mathcal{G}}}(\mathbf{H}^{(\ell)})$ with $\langle T_{V_{\mathcal{G}}}(\mathbf{H}^{(\ell)}), \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}^{(\ell)}) \rangle = 0$. Then, it is possible to learn a direction $T_{V_{\mathcal{G}}}(\mathbf{H}^{(\ell)})$ and a step size $\alpha_{\mathcal{G}}$ such that $\mathbf{D}$ is the Newton direction, $\mathbf{N} = (\nabla^2 V_{\mathcal{G}})^{-1} \nabla V_{\mathcal{G}}$.*

Proof. We aim to construct a direction $\mathbf{D} = \alpha_{\mathcal{G}}(\mathbf{H}) \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}) + T_{V_{\mathcal{G}}}(\mathbf{H})$ that matches the Newton direction:

$$\mathbf{N} = (\nabla_{\mathbf{H}}^2 V_{\mathcal{G}}(\mathbf{H}))^{-1} \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}).$$

Recall that by design, we have that $T_{V_{\mathcal{G}}}(\mathbf{H})$ is orthogonal to the energy gradient, i.e., $\langle T_{V_{\mathcal{G}}}(\mathbf{H}), \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}) \rangle = 0$. Then, we can express a Newton direction by the decomposition:

$$\mathbf{N} = \alpha_{\mathcal{G}}(\mathbf{H}) \nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}) + T_{V_{\mathcal{G}}}(\mathbf{H}).$$

Solving for the orthogonal component yields:

$$T_{V_G}(\mathbf{H}) = \mathbf{N} - \alpha_G(\mathbf{H}) \nabla_{\mathbf{H}} V_G(\mathbf{H}).$$

To enforce orthogonality, we require:

$$\langle \mathbf{N} - \alpha_G(\mathbf{H}) \nabla_{\mathbf{H}} V_G(\mathbf{H}), \nabla_{\mathbf{H}} V_G(\mathbf{H}) \rangle = 0.$$

Expanding and simplifying, we find:

$$\langle \mathbf{N}, \nabla_{\mathbf{H}} V_G(\mathbf{H}) \rangle - \alpha_G(\mathbf{H}) \|\nabla_{\mathbf{H}} V_G(\mathbf{H})\|^2 = 0,$$

and the optimal step size is given by:

$$\alpha_G(\mathbf{H}) = \frac{\langle \mathbf{N}, \nabla_{\mathbf{H}} V_G(\mathbf{H}) \rangle}{\|\nabla_{\mathbf{H}} V_G(\mathbf{H})\|^2},$$

showing that it is possible to learn a Newton direction, i.e., a quadratic energy convergence direction. $\square$

C. Complexity and Runtimes

Complexity. Each step of TANGO requires computing the gradient of the learned energy function $V_G(\mathbf{H}^{(\ell)})$, that is defined in Equation (8). This involves two main operations: (i) forward and backward passes through the energy network ENERGYGNN, which contains L_{energy} message-passing layers and an MLP; and (ii) automatic differentiation to compute $\nabla_{\mathbf{H}} V_G(\mathbf{H}^{(\ell)})$ with respect to the input node features. In parallel, the tangential flow direction $T_{V_G}(\mathbf{H}^{(\ell)})$ is obtained by projecting the vector field $\mathbf{M}^{(\ell)}$ computed by a separate TANGENTGNN with L_{tangent} layers onto the orthogonal complement of the energy gradient, as shown in Equation (3). This projection is of computational cost of $O(nd)$ per step, where $n = |\mathcal{V}|$ and d is the feature dimensionality. In addition, scalar coefficients α_G and β_G are computed from pooled node features using MLPs (Equations (9) and (11)). Assuming both ENERGYGNN and TANGENTGNN are message-passing architectures with linear complexity in the number of nodes and edges, and setting $L_{\text{energy}} = L_{\text{tangent}}$, the total complexity per layer becomes $O(L_{\text{gnn}} \cdot (n + m) \cdot d)$, where L_{gnn} is the number of GNN layers used in each subnetwork and $m = |\mathcal{E}|$ is the number of edges. Unrolling the dynamics over L steps, the overall computational complexity of TANGO is:

$$O(L \cdot L_{\text{gnn}} \cdot (|\mathcal{V}| + |\mathcal{E}|) \cdot d).$$

Memory. The memory footprint of TANGO is dominated by storing activations for backpropagation, as in any deep GNN. Using two subnetworks of the same backbone type (ENERGYGNN and TANGENTGNN) roughly doubles the number of feature tensors that need to be kept in memory. Nonetheless, we match the overall parameter budget to the underlying backbone by reducing widths where applicable (as shown in Table 9), so that the resulting models remain comparable in size. The asymptotic memory complexity remains linear in the number of nodes and edges and in the number of unrolled steps: the additional cost of computing $\nabla_{\mathbf{H}} V_G(\mathbf{H}^{(\ell)})$ is handled by standard automatic differentiation. In practice, we did not observe memory blow-up or numerical instabilities due to the projection in Equation (3).

Parameter count comparison. To ensure a fair comparison, we match the parameter budget of each backbone when instantiating TANGO. Table 4 reports parameter counts alongside mean performance and standard deviation across datasets and metrics. As shown, TANGO uses a comparable number of parameters to its corresponding backbones while achieving consistently stronger results. This protocol allows us to isolate the contribution of our Lyapunov-guided dynamics in TANGO from the number of parameters.

Runtimes. We benchmark training runtimes per iteration for TANGO instantiated on two backbones (GatedGCN and GPS) and compare against standard baselines across four datasets: Questions, Roman-Empire, ZINC-12k, and Diameter. The measurements are reported in Table 5. It is evident that TANGO introduces a moderate overhead relative to its corresponding backbone while remaining in the same order of magnitude as commonly used architectures. In particular, TANGO-GatedGCN is slower than GatedGCN but substantially faster than GPS-class methods, and TANGO-GPS scales proportionally with GPS. All measurements were taken under matched hyperparameters with 256 channels, 8 layers on a single NVIDIA RTX6000 Ada GPU with 48 GB memory.

Table 4. Comparison of models across datasets. Performance is reported as mean $\pm$ standard deviation, with the metric indicated; $\downarrow$ means lower is better and $\uparrow$ means higher is better.

Dataset	Model	Params	Performance (metric)
ZINC-12k	GatedGCN	503,013	0.282 $\pm$ 0.015 (MAE $\downarrow$)
	TANGO	503,409	0.128 $\pm$ 0.011 (MAE $\downarrow$)
	GPS	423,717	0.070 $\pm$ 0.004 (MAE $\downarrow$)
	TANGO	422,947	0.062 $\pm$ 0.005 (MAE $\downarrow$)
Roman-Empire	GatedGCN	541,086	74.46 $\pm$ 0.54 (Acc $\uparrow$)
	TANGO	520,822	91.89 $\pm$ 0.30 (Acc $\uparrow$)
	GPS	524,218	87.04 $\pm$ 0.58 (Acc $\uparrow$)
	TANGO	525,016	91.08 $\pm$ 0.57 (Acc $\uparrow$)
Peptides-func	GatedGCN	496,184	58.64 $\pm$ 0.77 (AP $\uparrow$)
	TANGO	496,590	68.92 $\pm$ 0.40 (AP $\uparrow$)
	GPS	504,362	65.35 $\pm$ 0.41 (AP $\uparrow$)
	TANGO	502,938	70.21 $\pm$ 0.43 (AP $\uparrow$)

Table 5. Training runtime comparison per epoch (ms) across datasets and baselines. TANGO achieves a similar runtime to other methods. We note that while TANGO requires more computation time than its backbone GNN, it remains efficient and within the same order of magnitude of computations as other methods, while offering improved performance as shown in the results in Tables 1, 4 and 11.

Model	Questions	Roman-Empire	ZINC-12k	Diameter
GIN	108.72	23.32	382.63	450.21
GCN	69.77	14.96	249.45	294.35
GatedGCN	129.92	27.86	453.76	537.57
GAT	112.40	24.12	398.02	471.40
GPS	429.08	92.08	1506.05	1822.03
GRIT	520.00	111.57	1865.06	2163.81
TANGO-GatedGCN	184.98	39.66	653.29	778.22
TANGO-GPS	694.27	148.96	2435.85	2899.24

D. Experimental Details

In this section, we provide additional experimental details.

Computational Resources. Our experiments are run on NVIDIA RTX6000 Ada with 48GB of memory. Our code is implemented in PyTorch (Paszke et al., 2019), and will be publicly released upon acceptance.

Baselines. Beyond methods cited directly in tables, we now enlist different classical and state-of-the-art GNN baselines. Specifically:

- Classical MPNNs, i.e., GCN (Kipf & Welling, 2017), GraphSAGE (Hamilton et al., 2017), GAT (Veličković et al., 2018), GatedGCN (Bresson & Laurent, 2018), GIN (Xu et al., 2019), GINE (Hu et al., 2020), GCNII (Chen et al., 2020), and CoGNN (Finkelshtein et al., 2024);
- Heterophily-specific models, i.e., H2GCN (Zhu et al., 2020), CPGNN (Zhu et al., 2021), FAGCN (Bo et al., 2021), GPR-GNN (Chien et al., 2021), FSGNN (Maurya et al., 2022), GloGNN (Li et al., 2022), GBK-GNN (Du et al., 2022), and JacobiConv (Wang & Zhang, 2022);
- DE-DGNs, i.e., DGC (Wang et al., 2021), GRAND (Chamberlain et al., 2021b), GraphCON (Rusch et al., 2022), A-DGN (Gravina et al., 2023), SWAN (Gravina et al., 2025), PH-DGN (Heilig et al., 2025);
- Graph Transformers, i.e., Transformer (Vaswani et al., 2017; Dwivedi & Bresson, 2021), GT (Shi et al., 2021), SAN (Kreuzer et al., 2021), GPS (Rampásek et al., 2022), GOAT (Kong et al., 2023), Exphormer (Shirzad et al., 2023), GRIT (Ma et al., 2023);

- Higher-Order DGNs, i.e., DIGL (Gasteiger et al., 2019), MixHop (Abu-El-Haija et al., 2019), and DRew (Gutteridge et al., 2023).
- SSM-based GNN, i.e., Graph-Mamba (Wang et al., 2024a) and GMN (Behrouz & Hashemi, 2024).
- Rewiring techniques, i.e., GRAND (Chamberlain et al., 2021b), DRew (Gutteridge et al., 2023).

D.1. Forward Euler discretization and stability

Recall that the continuous-time dynamics of TANGO are given by Equation (2) where V_G is a non-negative Lyapunov energy, $\nabla_{\mathbf{H}} V_G$ is its gradient with respect to node features, and T_{V_G} is constrained to be orthogonal to this gradient whenever $\nabla_{\mathbf{H}} V_G \neq 0$. In discrete depth, we implement TANGO using the forward Euler residual update as shown in Equation (5) for $\ell = 0, \dots, L-1$, with step size $\epsilon > 0$. In classical numerical analysis, explicit Euler schemes can be sensitive when applied to a *fixed* stiff ODE: for a given vector field, only a restricted range of step sizes yields stable trajectories (Ascher & Petzold, 1998). In TANGO, however, we do not discretize a predetermined ODE. Instead, the vector field is *learned*, while ϵ is selected from a small grid. Therefore, the discussion below should be interpreted as a local consistency statement relating the discrete update to the continuous-time Lyapunov dynamics, while empirical stability is assessed through training behavior and validation performance.

Our continuous-time analysis in Section 4 shows that, under the assumptions of Proposition 1, the energy $V_G(\mathbf{H}(t))$ is non-increasing along trajectories of Equation (2). The forward Euler update Equation (5) follows this dissipation law up to the usual second-order discretization term. Let

$$F(\mathbf{H}) = -\alpha_G(\mathbf{H})\nabla_{\mathbf{H}} V_G(\mathbf{H}) + \beta_G(\mathbf{H})T_{V_G}(\mathbf{H})$$

denote the vector field in Equation (2). Viewing $\mathbf{H}$ and $F(\mathbf{H})$ as vectors in $\mathbb{R}^{nd}$ equipped with the Frobenius norm, assume that V_G is twice continuously differentiable and that its Hessian is bounded in operator norm by L_V on the subset of feature space visited by the dynamics. A second-order Taylor expansion of V_G along the direction $\epsilon F(\mathbf{H}^{(\ell)})$ gives

$$\begin{aligned} V_G(\mathbf{H}^{(\ell+1)}) &= V_G(\mathbf{H}^{(\ell)}) + \epsilon \left\langle \nabla_{\mathbf{H}} V_G(\mathbf{H}^{(\ell)}), F(\mathbf{H}^{(\ell)}) \right\rangle_F \\ &\quad + \frac{\epsilon^2}{2} F(\mathbf{H}^{(\ell)})^\top \nabla_{\mathbf{H}}^2 V_G(\boldsymbol{\xi}^{(\ell)}) F(\mathbf{H}^{(\ell)}), \end{aligned} \quad (14)$$

for some intermediate point $\boldsymbol{\xi}^{(\ell)}$ on the line segment between $\mathbf{H}^{(\ell)}$ and $\mathbf{H}^{(\ell+1)}$. By construction, T_{V_G} is orthogonal to $\nabla_{\mathbf{H}} V_G$ whenever $\nabla_{\mathbf{H}} V_G \neq 0$, and $\alpha_G(\mathbf{H}^{(\ell)}) \geq 0$, so the first-order term satisfies

$$\left\langle \nabla_{\mathbf{H}} V_G(\mathbf{H}^{(\ell)}), F(\mathbf{H}^{(\ell)}) \right\rangle_F = -\alpha_G(\mathbf{H}^{(\ell)}) \|\nabla_{\mathbf{H}} V_G(\mathbf{H}^{(\ell)})\|_F^2 \leq 0.$$

Using the Hessian bound, the second-order remainder term is bounded above by

$$\frac{\epsilon^2}{2} F(\mathbf{H}^{(\ell)})^\top \nabla_{\mathbf{H}}^2 V_G(\boldsymbol{\xi}^{(\ell)}) F(\mathbf{H}^{(\ell)}) \leq \frac{L_V}{2} \epsilon^2 \|F(\mathbf{H}^{(\ell)})\|_F^2.$$

Combining these two observations, we obtain

$$V_G(\mathbf{H}^{(\ell+1)}) \leq V_G(\mathbf{H}^{(\ell)}) - \epsilon \alpha_G(\mathbf{H}^{(\ell)}) \|\nabla_{\mathbf{H}} V_G(\mathbf{H}^{(\ell)})\|_F^2 + \frac{L_V}{2} \epsilon^2 \|F(\mathbf{H}^{(\ell)})\|_F^2. \quad (15)$$

This estimate shows that the discrete-time dynamics follow the continuous-time Lyapunov dissipation law up to a second-order term in the step size. In particular, if $\|F(\mathbf{H}^{(\ell)})\|_F > 0$ and

$$\epsilon \leq \frac{2\alpha_G(\mathbf{H}^{(\ell)}) \|\nabla_{\mathbf{H}} V_G(\mathbf{H}^{(\ell)})\|_F^2}{L_V \|F(\mathbf{H}^{(\ell)})\|_F^2},$$

then the upper bound in Equation (15) is non-increasing. If $F(\mathbf{H}^{(\ell)}) = 0$, the Euler step leaves the features unchanged. If the numerator vanishes, for example when $\alpha_G(\mathbf{H}^{(\ell)}) = 0$ or $\nabla_{\mathbf{H}} V_G(\mathbf{H}^{(\ell)}) = 0$, this estimate does not guarantee finite-step energy decrease; it only implies that the possible energy change from the Euler discretization is second order in ϵ . Under

standard smoothness assumptions on the learned vector field, explicit Euler trajectories converge to the corresponding continuous-time trajectories on fixed finite time intervals as $\epsilon \rightarrow 0$, with the final integration time held fixed. In this limit, the continuous-time Lyapunov property is recovered: the energy is non-increasing, and it is strictly decreasing only at states where $\alpha_G(\mathbf{H}) > 0$ and $\|\nabla_{\mathbf{H}} V_G(\mathbf{H})\|_F > 0$. In practice, we select ϵ from a range where $V_G(\mathbf{H}^{(\ell)})$ is empirically non-increasing along depth and training remains stable.

This behavior is analogous to what is observed in residual networks and continuous-depth models. Architectures such as Neural Ordinary Differential Equations for Euclidean data and graph-based ODE models such as in (Chamberlain et al., 2021b; Eliasof et al., 2021; Gravina et al., 2023; Rusch et al., 2022; Choi et al., 2023) are often implemented using explicit Euler or related explicit Runge–Kutta schemes on a *learned* vector field. In such models, the practical stability of the discretization is governed by the interaction between the learned dynamics, the chosen step size or solver, and the training objective. Our contribution is orthogonal to the specific choice of time integrator. TANGO proposes a Lyapunov-structured vector field on node embeddings, decomposed into an energy-dissipating gradient term and an energy-tangential term, and instantiates it with the same forward Euler step that is standard in many ODE-inspired GNNs (Chamberlain et al., 2021b; Eliasof et al., 2021; Gravina et al., 2023). Other integrators, such as implicit-explicit schemes, semi-implicit methods, or higher-order Runge–Kutta methods, could in principle be combined with TANGO as well, and a comparison of such integrators in this setting is an interesting avenue for future work.

D.2. Synthetic Example

In the synthetic example in Figure 2, we demonstrate the effectiveness of TANGO on a bottleneck-sensitive propagation task. We consider a Barbell graph where all node features are set to 0 except for the left-most node, which is set to 1, as shown in Figure 2(a). The goal is to propagate this information through the bottleneck and across the graph. A Dirichlet-energy gradient flow with 50 layers struggles to transmit the signal to the right clique, as shown in Figure 2(b). In contrast, TANGO, which combines energy descent with tangential flow, propagates the signal more effectively across the bottleneck, as shown in Figure 2(c).

Bottleneck and feature-propagation diagnostics. To make the oversquashing connection more direct, we further evaluate the same barbell topology using structural and sensitivity-based diagnostics. We measure the structural severity of the bottleneck by effective resistance, following topology-based analyses of oversquashing (Black et al., 2023), and measure feature propagation using Jacobian responses, following sensitivity-based views of bottlenecks (Topping et al., 2022; Di Giovanni et al., 2023). Effective resistance confirms that cross-clique pairs are the hardest communication regime: mean resistance is 0.20 for same-clique pairs, 1.18 for bridge pairs, and 2.36 for cross-clique pairs.

Budget	Metric	TANGO	Gradient flow	GCN	Gain/GF	Gain/GCN
50 steps	CCT $\uparrow$	0.0508	0.0237	0.0172	+114.3%	+195.3%
50 steps	CJR $\uparrow$	0.0745	0.0241	0.0176	+209.1%	+323.3%
50 steps	SBR $\uparrow$	1.1229	0.3238	0.2174	+246.8%	+416.6%
100 steps	CCT $\uparrow$	0.0564	0.02495	0.01839	+126.1%	+206.7%
100 steps	CJR $\uparrow$	0.1370	0.0366	0.0292	+274.3%	+369.2%
100 steps	SBR $\uparrow$	1.1487	0.3312	0.2217	+246.8%	+418.1%

Table 6. Bottleneck propagation diagnostics on the barbell graph. CCT denotes cross-clique transmission, CJR denotes mean absolute cross-clique Jacobian response, and SBR denotes the selected/background Jacobian ratio. Higher is better for all metrics; gain columns report the relative improvement of TANGO over the corresponding baseline.

The diagnostic results in Table 6 show that TANGO obtains the strongest propagation on the highest-resistance cross-clique pairs under both 50-step and 100-step budgets. Relative to the gradient-flow baseline, the gains range from +114.3% to +274.3% across CCT and CJR and are +246.8% for SBR under both budgets; the corresponding gains over GCN are even larger. Thus, the benefit is visible not only in downstream benchmark scores, but also in direct measures of cross-bottleneck feature transmission and source-target sensitivity.

Controlled one-step analysis of the tangential term. To isolate the tangential component from unrelated architectural factors, we also compare matched one-step updates from the same visited states on the barbell graph. For each state, we compare a gradient-only update with the full TANGO update and record the alignment of the learned tangential direction with a loss-reducing tangent direction, the change in task loss, and the change in cross-clique transmission.

Regime	$\cos(\mathbf{d}_{\text{tan}}, \mathbf{g}_{\text{tan}}^{\mathcal{L}}) \uparrow$	$\Delta\text{Loss GF} \downarrow$	$\Delta\text{Loss TANGO} \downarrow$	$\Delta\text{Transmission GF} \uparrow$	$\Delta\text{Transmission TANGO} \uparrow$
Flat-energy states	0.2857	-0.0005	-0.0904	+0.0006	+0.1105
Low-loss states	0.2339	-0.00003	-0.0103	+0.00005	+0.0131

Table 7. Controlled comparison on the barbell graph. GF denotes the gradient-flow update, and $\cos(\mathbf{d}_{\text{tan}}, \mathbf{g}_{\text{tan}}^{\mathcal{L}})$ measures alignment between the learned tangential direction and a loss-reducing tangent direction. The full TANGO update gives larger loss reduction and cross-clique transmission gains from the same states.

This controlled comparison supports the role assigned to the tangential component in Section 4. In flat-energy states, where pure gradient descent makes almost no progress, the learned tangential direction provides a useful loss-reducing and transmission-enhancing update. In low-loss states, the update remains beneficial rather than pulling the representation away from a good solution.

D.3. Graph Property Prediction

Dataset. We construct our benchmark following the protocol introduced by Gravina et al. (2023). Graph instances are synthetically generated from a variety of canonical topologies, including Erdős–Rényi, Barabasi-Albert, caveman, tree, and grid models. Each graph consists of 25 to 35 nodes, with node features initialized as random identifiers sampled uniformly from the interval $[0, 1)$. The prediction targets encompass several structural tasks: computing the shortest paths from a source node, estimating node eccentricity, and determining graph diameter. The complete dataset contains 7,040 graphs, split into 5,120 for training, 640 for validation, and 1,280 for testing. These tasks inherently require capturing long-range dependencies, as they involve global graph computations such as shortest-path inference. As highlighted by Gravina et al. (2023), traditional algorithms such as Bellman-Ford and Dijkstra’s method require multiple rounds of message propagation, underscoring the need for expressive graph models. The benchmark graph families, such as caveman, tree, line, star, caterpillar, and lobster, frequently include structural bottlenecks that are known to induce oversquashing effects (Topping et al., 2022), posing additional challenges for message-passing-based GNNs.

Experimental Setup. We adopt the same evaluation framework as Gravina et al. (2023), including datasets, training routines, and hyperparameter spaces. Model training is conducted using the Adam optimizer for up to 1500 epochs, with early stopping triggered after 100 consecutive epochs without improvement in the validation Mean Squared Error (MSE). Hyperparameters are selected via grid search, and performance is averaged over 4 independent runs with different random seeds for weight initialization. A summary of the hyperparameter grid used in our experiments is provided in Table 9.

D.4. Graph Benchmarks

Dataset. To comprehensively assess the capabilities of TANGO, we evaluate its performance on a diverse set of graph learning benchmarks curated by Dwivedi et al. (2023). The benchmark suite includes: *ZINC-12k*, a molecular regression dataset containing chemical compounds, where the goal is to predict the constrained solubility of each molecule. Graphs represent molecular structures, with atoms as nodes and chemical bonds as edges. Node and edge features encode atom types and bond types, respectively. *MNIST* and *CIFAR-10* superpixels are graph-structured versions of standard image classification datasets, where images are converted into sparse graphs of superpixels. Each superpixel forms a node, and edges are based on spatial adjacency. The tasks involve classifying digits (MNIST) and natural objects (CIFAR-10) based on graph-structured representations. *CLUSTER* and *PATTERN* are synthetic datasets designed to assess the relational inductive biases of graph neural networks. Both datasets are generated from stochastic block models (SBMs). In *CLUSTER*, the task is to group nodes by community, while *PATTERN* involves identifying specific structural patterns within each graph. These datasets span a variety of domains: chemical, image, and synthetic graphs, and are commonly used to benchmark architectural innovations in GNNs (Ma et al., 2023). We follow the official training, validation, and test splits provided by Dwivedi et al. (2023), ensuring consistency in evaluation across models.

Experimental Setup. We adhere to the training and evaluation protocol established in Dwivedi et al. (2023). For each dataset, we perform hyperparameter tuning via grid search, optimizing the corresponding evaluation metrics: Mean Absolute Error (MAE) for *ZINC-12k*, and classification accuracy for the remaining tasks. We use the AdamW optimizer and train all models for up to 300 epochs, with early stopping based on validation performance. To ensure comparability with prior work, we respect the same parameter budgets used in the original benchmark and maintain the architectural constraints defined for

fair evaluation. Each configuration is trained with three random seeds, and we report the average and standard deviation of the results. Hyperparameter ranges used in this set of experiments are summarized in Table 9.

D.5. Long Range Graph Benchmark

Dataset. To evaluate model performance on real-world graphs with significant long-range dependencies, we utilize the *Peptides-func* and *Peptides-struct* benchmarks introduced in Dwivedi et al. (2022b). These datasets represent peptide molecules as graphs, where nodes correspond to heavy (non-hydrogen) atoms, and edges denote chemical bonds. *Peptides-func* is a multi-label classification task with 10 functional categories, including antibacterial, antiviral, and signaling-related properties. In contrast, *Peptides-struct* focuses on regression, targeting physical and geometric attributes such as molecular inertia (weighted by atomic mass and valence), extreme atom-pair distances, sphericity, and average deviation from a best-fit plane. Together, the two datasets comprise 15,535 peptide graphs and roughly 2.3 million nodes. We adopt the official train/validation/test partitions from Dwivedi et al. (2022b) and report mean and standard deviation across three different random seeds for each experiment.

Experimental Setup. We follow the evaluation protocol established in Dwivedi et al. (2022b), including dataset usage, training strategy, and model capacity constraints. Hyperparameter tuning is performed via grid search, optimizing Average Precision (AP) for the classification task and Mean Absolute Error (MAE) for the regression task. All models are trained using the AdamW optimizer for up to 300 epochs, with early stopping based on validation performance. To ensure fairness and comparability, all models adhere to the 500K parameter limit, in line with the settings of Dwivedi et al. (2022b) and Gutteridge et al. (2023). Each configuration is run three times with different weight initializations, and the results are averaged. Details of the hyperparameter ranges considered are shown in Table 9.

D.6. Heterophilic Node Classification

Dataset. For evaluating performance in heterophilic graph settings, we consider five benchmark tasks introduced by Platonov et al. (2023): *Roman-Empire*, *Amazon-Ratings*, *Minesweeper*, *Tolokers*, and *Questions*. These datasets span a diverse range of domains and graph topologies. *Roman-Empire* is constructed from the Wikipedia article on the Roman Empire, where nodes represent words and edges capture either sequential adjacency or syntactic relations. The task is node classification with 18 syntactic categories, and the underlying graph is sparse and chain-structured, suggesting the presence of long-range dependencies. *Amazon-Ratings* originates from Amazon’s product co-purchasing graph. Nodes correspond to products, and they are linked if they are frequently bought together. The classification task involves predicting discretized average product ratings (five classes), with node features derived from fastText embeddings of product descriptions. *Minesweeper* is a synthetic dataset modeled as a 100×100 grid. Nodes represent individual cells, with edges connecting adjacent cells. A random 20% of nodes are labeled as mines, and the objective is to classify mine-containing cells based on one-hot features that encode the number of neighboring mines. *Tolokers* is based on the Toloka crowdsourcing platform (Likhobaba et al., 2023), where each node is a worker (toloker), and edges indicate co-participation on the same project. The task involves binary classification to detect whether a worker has been banned, using node features from user profiles and performance metrics. *Questions* draws from user interaction data on Yandex Q, a question-answering forum. Nodes represent users, and edges capture answering interactions. The goal is to identify active users, using input features derived from user-provided descriptions. A summary of dataset statistics is provided in Table 8.

Table 8. Statistics of the heterophilic node classification datasets.

	Roman-empire	Amazon-ratings	Minesweeper	Tolokers	Questions
N. nodes	22,662	24,492	10,000	11,758	48,921
N. edges	32,927	93,050	39,402	519,000	153,540
Avg degree	2.91	7.60	7.88	88.28	6.28
Diameter	6,824	46	99	11	16
Node features	300	300	7	10	301
Classes	18	5	2	2	2
Edge homophily	0.05	0.38	0.68	0.59	0.84

Experimental Setup. Our experimental procedure aligns with that of Freitas et al. (2021) and Platonov et al. (2023). We conduct a grid search to optimize model performance, using classification accuracy for the *Roman-Empire* and *Amazon-Ratings* tasks, and ROC-AUC for the *Minesweeper*, *Tolokers*, and *Questions* tasks. Each model is trained with the AdamW optimizer for up to 300 epochs. Our experiments follow the official dataset splits provided by Platonov et al. (2023). For

each model configuration, we perform multiple training runs with different random seeds and report the mean and standard deviation of the results. The hyperparameter grid explored in these experiments is summarized in Table 9.

D.7. Hyperparameters

In Table 9, we summarize the hyperparameter grids used for tuning our TANGO across different benchmarks. In particular, we have followed similar practices from the literature (Gravina et al., 2025; Rusch et al., 2022). Alongside standard training hyperparameters such as learning rate, weight decay, and batch size, our method introduces several additional components. These include the number of unrolled steps L (corresponding to the depth of the energy-based dynamics), the hidden dimension d of node features, and the number of message-passing layers L_{gnn} used within the internal ENERGYGNN and TANGENTGNN modules. In all experiments, we share the architecture depth between ENERGYGNN and TANGENTGNN. We also tune the step size ϵ used in the forward Euler update (Equation (5)), which controls the integration scale of the continuous dynamics. We explore multiple values of L to assess how the number of dynamical steps impacts long-range propagation across different tasks. Details of the complete hyperparameter grid can be found in Table 9.

Table 9. Hyperparameter grids used during model selection for the different benchmark categories: *GraphPropPred* (Diameter, SSSP, Eccentricity), *LRGB* (Peptides-func/struct), *Graph Benchmarks* (ZINC-12k, MNIST, CIFAR-10, CLUSTER, PATTERN), and *Node Classification* (Roman-Empire, Amazon-Ratings, Minesweeper, Tolokers, Questions).

Hyperparameter	<i>GraphPropPred</i>	<i>LRGB</i>	<i>Graph Benchmarks</i>	<i>Node Classification</i>
Unrolled steps L	{1,5,10,20}	{2,4,8,16,32}	{2,4,8,16,32}	{2,4,8,16,32}
GNN layers L_{gnn}	{1,2,4,8,16}	{1,2,4,8,16}	{1,2,4,8,16}	{1,2,4,8,16}
Feature dimension d	{10, 20, 30}	{64, 128, 256}	{64, 128, 256}	{64, 128, 256}
Step size ϵ	{0.001, 0.1, 1.0}	{0.001, 0.1, 1.0}	{0.001, 0.1, 1.0}	{0.001, 0.1, 1.0}
Learning rate	{1e-3, 1e-4}	{1e-3, 1e-4}	{1e-3, 1e-4}	{1e-3, 1e-4}
Weight decay	{0, 1e-6, 1e-5}	{0, 1e-6, 1e-5}	{0, 1e-6, 1e-5}	{0, 1e-6, 1e-5}
Activation function (σ)	ReLU	ELU, GELU, ReLU	ELU, GELU, ReLU	ELU, GELU, ReLU
Batch size	{32, 64, 128}	{32, 64, 128}	{32, 64, 128}	N/A

E. Additional Results and Comparisons

E.1. Heterophilic Node Classification

We report and compare the performance of our TANGO with other recent benchmarks on the heterophilic node classification datasets from Platonov et al. (2023), in Table 11. As can be seen from the Table, TANGO offers strong performance that is similar or better than recent state-of-the-art methods, further demonstrating its effectiveness.

E.2. Ablation on Depth: Number of Layers

Setup. We study the effect of depth by varying the number of layers and measuring downstream performance on ROMAN-EMPIRE. All runs use identical training settings and data splits; only the depth differs.

Results. Table 10 shows that TANGO benefits from increased depth up to a task-dependent plateau. For TANGO-GatedGCN, performance improves steadily and saturates around 16 layers. For TANGO-GPS, gains persist up to 8 to 16 layers and then flatten. Importantly, we do not observe degradation when adding more layers within the explored range.

Table 10. Ablation on the number of layers for ROMAN-EMPIRE. Values are mean classification accuracy (%) $\pm$ standard deviation.

Layers	2	4	8	16	32
TANGO-GatedGCN	87.13 $\pm$ 0.36	89.08 $\pm$ 0.41	90.80 $\pm$ 0.37	91.89 $\pm$ 0.30	91.82 $\pm$ 0.44
TANGO-GPS	86.98 $\pm$ 0.48	88.71 $\pm$ 0.59	91.08 $\pm$ 0.57	91.01 $\pm$ 0.64	91.05 $\pm$ 0.60

E.3. Additional Comparisons

The comparisons in Section 5 provide a focused comparison with directly related methods and baseline backbones. In addition, we now provide a more comprehensive comparison in Table 14 and Table 15 to facilitate a more thorough comparison with recent methods. As shown, our TANGO also performs well in these comparisons.

Table 11. Mean test set score and std averaged over the splits from Platonov et al. (2023). **First**, **second**, and **third** best results for each task are color-coded. We mark each method once; if two variants are among the leading methods, we mark the best-performing variant.

Model	Roman-empire Acc $\uparrow$	Amazon-ratings Acc $\uparrow$	Minesweeper AUC $\uparrow$	Tolokers AUC $\uparrow$	Questions AUC $\uparrow$
MPNNs					
GIN	72.82 $\pm$ 0.58	46.96 $\pm$ 0.44	88.04 $\pm$ 0.78	81.79 $\pm$ 0.55	75.90 $\pm$ 1.03
GAT	80.87 $\pm$ 0.30	49.09 $\pm$ 0.63	92.01 $\pm$ 0.68	83.70 $\pm$ 0.47	77.43 $\pm$ 1.20
GAT-sep	88.75 $\pm$ 0.41	52.70 $\pm$ 0.62	93.91 $\pm$ 0.35	83.78 $\pm$ 0.43	76.79 $\pm$ 0.71
Gated-GCN	74.46 $\pm$ 0.54	43.00 $\pm$ 0.32	87.54 $\pm$ 1.22	77.31 $\pm$ 1.14	76.61 $\pm$ 1.13
GCN	73.69 $\pm$ 0.74	48.70 $\pm$ 0.63	89.75 $\pm$ 0.52	83.64 $\pm$ 0.67	76.09 $\pm$ 1.27
CO-GNN(Σ , Σ)	91.57 $\pm$ 0.32	51.28 $\pm$ 0.56	95.09 $\pm$ 1.18	83.36 $\pm$ 0.89	80.02 $\pm$ 0.86
CO-GNN(μ , μ)	91.37 $\pm$ 0.35	54.17 $\pm$ 0.37	97.31 $\pm$ 0.41	84.45 $\pm$ 1.17	76.54 $\pm$ 0.95
SAGE	85.74 $\pm$ 0.67	53.63 $\pm$ 0.39	93.51 $\pm$ 0.57	82.43 $\pm$ 0.44	76.44 $\pm$ 0.62
Graph Transformers					
Expformer	89.03 $\pm$ 0.37	53.51 $\pm$ 0.46	90.74 $\pm$ 0.53	83.77 $\pm$ 0.78	73.94 $\pm$ 1.06
NAGphormer	74.34 $\pm$ 0.77	51.26 $\pm$ 0.72	84.19 $\pm$ 0.66	78.32 $\pm$ 0.95	68.17 $\pm$ 1.53
GOAT	71.59 $\pm$ 1.25	44.61 $\pm$ 0.50	81.09 $\pm$ 1.02	83.11 $\pm$ 1.04	75.76 $\pm$ 1.66
GPS _{GAT+Performer} (RWSE)	87.04 $\pm$ 0.58	49.92 $\pm$ 0.68	91.08 $\pm$ 0.58	84.38 $\pm$ 0.91	77.14 $\pm$ 1.49
GT	86.51 $\pm$ 0.73	51.17 $\pm$ 0.66	91.85 $\pm$ 0.76	83.23 $\pm$ 0.64	77.95 $\pm$ 0.68
GT-sep	87.32 $\pm$ 0.39	52.18 $\pm$ 0.80	92.29 $\pm$ 0.47	82.52 $\pm$ 0.92	78.05 $\pm$ 0.93
Heterophily-Designated GNNs					
FAGCN	65.22 $\pm$ 0.56	44.12 $\pm$ 0.30	88.17 $\pm$ 0.73	77.75 $\pm$ 1.05	77.24 $\pm$ 1.26
FSGNN	79.92 $\pm$ 0.56	52.74 $\pm$ 0.83	90.08 $\pm$ 0.70	82.76 $\pm$ 0.61	78.86 $\pm$ 0.92
GBK-GNN	74.57 $\pm$ 0.47	45.98 $\pm$ 0.71	90.85 $\pm$ 0.58	81.01 $\pm$ 0.67	74.47 $\pm$ 0.86
GloGNN	59.63 $\pm$ 0.69	36.89 $\pm$ 0.14	51.08 $\pm$ 1.23	73.39 $\pm$ 1.17	65.74 $\pm$ 1.19
GPR-GNN	64.85 $\pm$ 0.27	44.88 $\pm$ 0.34	86.24 $\pm$ 0.61	72.94 $\pm$ 0.97	55.48 $\pm$ 0.91
JacobiConv	71.14 $\pm$ 0.42	43.55 $\pm$ 0.48	89.66 $\pm$ 0.40	68.66 $\pm$ 0.65	73.88 $\pm$ 1.16
Ours					
TANGO _{GCN}	89.67 $\pm$ 0.68	52.98 $\pm$ 0.71	98.37 $\pm$ 0.49	85.57 $\pm$ 0.73	79.86 $\pm$ 1.14
TANGO _{GIN}	89.19 $\pm$ 0.62	50.76 $\pm$ 0.47	97.38 $\pm$ 0.50	84.39 $\pm$ 0.61	78.84 $\pm$ 0.96
TANGO _{GatedGCN}	91.89 $\pm$ 0.30	52.60 $\pm$ 0.53	98.32 $\pm$ 0.59	85.51 $\pm$ 0.98	80.39 $\pm$ 1.04
TANGO _{GPS}	91.08 $\pm$ 0.57	53.83 $\pm$ 0.32	98.39 $\pm$ 0.54	85.66 $\pm$ 1.01	80.32 $\pm$ 1.07

E.4. Ablation Study

Setup. We conduct two key ablation studies to better understand the contributions of the energy function and the tangential flow in TANGO. Specifically, we aim to answer the following questions:

- (i) Does downstream performance benefit from incorporating a tangential term even when the underlying GNN is not the gradient of an energy function?
- (ii) Is the observed improvement due to the tangential nature of the added component, or simply due to additional parameters and network?

To address these questions, we design two controlled experiments. For comprehensive coverage, we evaluate one representative dataset from each benchmark group: ZINC-12k, Roman-empire, Peptides-func, and Diameter. All experiments are run with two backbone architectures, GatedGCN and GPS. For reference, we also report the performance of the original backbones.

Results. For ablation (i), we compare TANGO against a variant we call TANGO-NON-ENERGY, in which the gradient-based energy descent term $\nabla_{\mathbf{H}} V_{\mathcal{G}}(\mathbf{H}^{(\ell)})$ in Equation (5) is replaced by intermediate node features from the same GNN backbone, as detailed in Equation (6). These features are computed using the same architecture but are not guaranteed to correspond to the gradient of any scalar energy function. This setup ensures fairness in capacity while removing the energy-based structure. As shown in Table 12, although both variants benefit from the inclusion of the tangential component, the full TANGO consistently outperforms TANGO-NON-ENERGY, confirming that leveraging a valid energy gradient contributes meaningfully to downstream performance.

For ablation (ii), we isolate the effect of the tangential nature of the added direction. In this variant, denoted TANGO-NON-TANGENT, we use the same output from the tangential network as in Equation (10) but omit the orthogonal projection

Table 12. Ablation study on the importance of using a gradient of an energy term in Equation (5).

Model	ZINC-12k MAE ↓	Roman-empire Acc. ↑	Peptides-func AP ↑	Diameter $\log_{10}(\text{MSE})$ ↓
GatedGCN	0.282 $\pm$ 0.015	74.46 $\pm$ 0.54	58.64 $\pm$ 0.77	0.1348 $\pm$ 0.0397
TANGO-NON-ENERGY _{GatedGCN}	0.138 $\pm$ 0.014	86.94 $\pm$ 0.43	68.07 $\pm$ 0.45	-0.5992 $\pm$ 0.0831
TANGO _{GatedGCN}	0.128 $\pm$ 0.011	91.89 $\pm$ 0.30	68.92 $\pm$ 0.40	-0.6681 $\pm$ 0.0745
GPS	0.070 $\pm$ 0.004	87.04 $\pm$ 0.58	65.35 $\pm$ 0.41	-0.5121 $\pm$ 0.0426
TANGO-NON-ENERGY _{GPS}	0.067 $\pm$ 0.004	89.00 $\pm$ 0.61	67.58 $\pm$ 0.39	-0.7178 $\pm$ 0.0729
TANGO _{GPS}	0.062 $\pm$ 0.005	91.08 $\pm$ 0.57	70.21 $\pm$ 0.43	-0.9772 $\pm$ 0.0518

Table 13. The importance of using a tangential term to the energy term in Equation (5).

Model	ZINC-12k MAE ↓	Roman-empire Acc. ↑	Peptides-func AP ↑	Diameter $\log_{10}(\text{MSE})$ ↓
GatedGCN	0.282 $\pm$ 0.015	74.46 $\pm$ 0.54	58.64 $\pm$ 0.77	0.1348 $\pm$ 0.0397
TANGO-NON-TANGENT _{GatedGCN}	0.186 $\pm$ 0.016	83.59 $\pm$ 0.48	68.01 $\pm$ 0.52	-0.2193 $\pm$ 0.0899
TANGO _{GatedGCN}	0.128 $\pm$ 0.011	91.89 $\pm$ 0.30	68.92 $\pm$ 0.40	-0.6681 $\pm$ 0.0745
GPS	0.070 $\pm$ 0.004	87.04 $\pm$ 0.58	65.35 $\pm$ 0.41	-0.5121 $\pm$ 0.0426
TANGO-NON-TANGENT _{GPS}	0.066 $\pm$ 0.010	88.57 $\pm$ 0.72	67.33 $\pm$ 0.59	-0.2916 $\pm$ 0.0404
TANGO _{GPS}	0.062 $\pm$ 0.005	91.08 $\pm$ 0.57	70.21 $\pm$ 0.43	-0.9772 $\pm$ 0.0518

step defined in Equation (3). Thus, while we still introduce an additional GNN term into the dynamics, it is not explicitly orthogonal to the energy gradient. Our results in Table 13 show that while this variant improves the performance compared with the baseline backbone, it also results in a drop in performance compared to the full TANGO. This highlights the importance of the tangential constraint, and its contribution towards improving the utilization of the learned energy function, as discussed in Section 4. Together, these ablations underscore the importance of both components in our design: (i) the principled learned energy descent, and (ii) the structured tangential update, as crucial for effective and flexible feature evolution.

Table 14. Results for Peptides-func and Peptides-struct averaged over 3 training seeds. Baseline results are taken from (Dwivedi et al., 2022b) and (Gutteridge et al., 2023). Re-evaluated methods employ the 3-layer MLP readout proposed in (Tönshoff et al., 2023a). Note that all MPNN-based methods include structural and positional encoding. [‡] means 3-layer MLP readout and residual connections are employed based on (Tönshoff et al., 2023a). This table is an extended version of the focused Table 3.

Model	Peptides-func AP $\uparrow$	Peptides-struct MAE $\downarrow$
MPNNs		
GCN	59.30 $\pm$ 0.23	0.3496 $\pm$ 0.0013
GINE	54.98 $\pm$ 0.79	0.3547 $\pm$ 0.0045
GCNII	55.43 $\pm$ 0.78	0.3471 $\pm$ 0.0010
GatedGCN	58.64 $\pm$ 0.77	0.3420 $\pm$ 0.0013
Multi-hop GNNs		
DIGL+MPNN	64.69 $\pm$ 0.19	0.3173 $\pm$ 0.0007
DIGL+MPNN+LapPE	68.30 $\pm$ 0.26	0.2616 $\pm$ 0.0018
MixHop-GCN	65.92 $\pm$ 0.36	0.2921 $\pm$ 0.0023
MixHop-GCN+LapPE	68.43 $\pm$ 0.49	0.2614 $\pm$ 0.0023
DRew-GCN	69.96 $\pm$ 0.76	0.2781 $\pm$ 0.0028
DRew-GCN+LapPE	71.50 $\pm$ 0.44	0.2536 $\pm$ 0.0015
DRew-GIN	69.40 $\pm$ 0.74	0.2799 $\pm$ 0.0016
DRew-GIN+LapPE	71.26 $\pm$ 0.45	0.2606 $\pm$ 0.0014
DRew-GatedGCN	67.33 $\pm$ 0.94	0.2699 $\pm$ 0.0018
DRew-GatedGCN+LapPE	69.77 $\pm$ 0.26	0.2539 $\pm$ 0.0007
Transformers		
Transformer+LapPE	63.26 $\pm$ 1.26	0.2529 $\pm$ 0.0016
SAN+LapPE	63.84 $\pm$ 1.21	0.2683 $\pm$ 0.0043
GraphGPS+LapPE	65.35 $\pm$ 0.41	0.2500 $\pm$ 0.0005
Modified and Re-evaluated[‡]		
GCN	68.60 $\pm$ 0.50	0.2460 $\pm$ 0.0007
GINE	66.21 $\pm$ 0.67	0.2473 $\pm$ 0.0017
GatedGCN	67.65 $\pm$ 0.47	0.2477 $\pm$ 0.0009
GraphGPS	65.34 $\pm$ 0.91	0.2509 $\pm$ 0.0014
DE-GNNs		
GRAND	57.89 $\pm$ 0.62	0.3418 $\pm$ 0.0015
GraphCON	60.22 $\pm$ 0.68	0.2778 $\pm$ 0.0018
A-DGN	59.75 $\pm$ 0.44	0.2874 $\pm$ 0.0021
SWAN	67.51 $\pm$ 0.39	0.2485 $\pm$ 0.0009
Graph SSMs		
Graph-Mamba	67.39 $\pm$ 0.87	0.2478 $\pm$ 0.0016
GMN	70.71 $\pm$ 0.83	0.2473 $\pm$ 0.0025
Ours		
TANGO _{GCN}	69.17 $\pm$ 0.31	0.2432 $\pm$ 0.0011
TANGO _{GIN}	68.78 $\pm$ 0.66	0.2440 $\pm$ 0.0024
TANGO _{GATEDGCN}	68.92 $\pm$ 0.40	0.2451 $\pm$ 0.0006
TANGO _{GPS}	70.21 $\pm$ 0.43	0.2422 $\pm$ 0.0014

Table 15. Mean test set score and std averaged over the splits from Platonov et al. (2023). This table is an extended version of the focused Table 11. Baseline results are reported from (Finkelshtein et al., 2024; Platonov et al., 2023; Müller et al., 2024; Luan et al., 2024).

Model	Roman-empire Acc $\uparrow$	Amazon-ratings Acc $\uparrow$	Minesweeper AUC $\uparrow$	Tolokers AUC $\uparrow$	Questions AUC $\uparrow$
MPNNs					
GIN	72.82 $\pm$ 0.58	46.96 $\pm$ 0.44	88.04 $\pm$ 0.78	81.79 $\pm$ 0.55	75.90 $\pm$ 1.03
GAT	80.87 $\pm$ 0.30	49.09 $\pm$ 0.63	92.01 $\pm$ 0.68	83.70 $\pm$ 0.47	77.43 $\pm$ 1.20
GAT-sep	88.75 $\pm$ 0.41	52.70 $\pm$ 0.62	93.91 $\pm$ 0.35	83.78 $\pm$ 0.43	76.79 $\pm$ 0.71
GAT (LapPE)	84.80 $\pm$ 0.46	44.90 $\pm$ 0.73	93.50 $\pm$ 0.54	84.99 $\pm$ 0.54	76.55 $\pm$ 0.84
GAT (RWSE)	86.62 $\pm$ 0.53	48.58 $\pm$ 0.41	92.53 $\pm$ 0.65	85.02 $\pm$ 0.67	77.83 $\pm$ 1.22
GAT (DEG)	85.51 $\pm$ 0.56	51.65 $\pm$ 0.60	93.04 $\pm$ 0.62	84.22 $\pm$ 0.81	77.10 $\pm$ 1.23
Gated-GCN	74.46 $\pm$ 0.54	43.00 $\pm$ 0.32	87.54 $\pm$ 1.22	77.31 $\pm$ 1.14	76.61 $\pm$ 1.13
GCN	73.69 $\pm$ 0.74	48.70 $\pm$ 0.63	89.75 $\pm$ 0.52	83.64 $\pm$ 0.67	76.09 $\pm$ 1.27
GCN (LapPE)	83.37 $\pm$ 0.55	44.35 $\pm$ 0.36	94.26 $\pm$ 0.49	84.95 $\pm$ 0.78	77.79 $\pm$ 1.34
GCN (RWSE)	84.84 $\pm$ 0.55	46.40 $\pm$ 0.55	93.84 $\pm$ 0.48	85.11 $\pm$ 0.77	77.81 $\pm$ 1.40
GCN (DEG)	84.21 $\pm$ 0.47	50.01 $\pm$ 0.69	94.14 $\pm$ 0.50	82.51 $\pm$ 0.83	76.96 $\pm$ 1.21
CO-GNN(Σ , Σ)	91.57 $\pm$ 0.32	51.28 $\pm$ 0.56	95.09 $\pm$ 1.18	83.36 $\pm$ 0.89	80.02 $\pm$ 0.86
CO-GNN(μ , μ)	91.37 $\pm$ 0.35	54.17 $\pm$ 0.37	97.31 $\pm$ 0.41	84.45 $\pm$ 1.17	76.54 $\pm$ 0.95
SAGE	85.74 $\pm$ 0.67	53.63 $\pm$ 0.39	93.51 $\pm$ 0.57	82.43 $\pm$ 0.44	76.44 $\pm$ 0.62
Graph Transformers					
Exphormer	89.03 $\pm$ 0.37	53.51 $\pm$ 0.46	90.74 $\pm$ 0.53	83.77 $\pm$ 0.78	73.94 $\pm$ 1.06
NAGphormer	74.34 $\pm$ 0.77	51.26 $\pm$ 0.72	84.19 $\pm$ 0.66	78.32 $\pm$ 0.95	68.17 $\pm$ 1.53
GOAT	71.59 $\pm$ 1.25	44.61 $\pm$ 0.50	81.09 $\pm$ 1.02	83.11 $\pm$ 1.04	75.76 $\pm$ 1.66
GPS	82.00 $\pm$ 0.61	53.10 $\pm$ 0.42	90.63 $\pm$ 0.67	83.71 $\pm$ 0.48	71.73 $\pm$ 1.47
GPS _{GCN} +Performer (LapPE)	83.96 $\pm$ 0.53	48.20 $\pm$ 0.67	93.85 $\pm$ 0.41	84.72 $\pm$ 0.77	77.85 $\pm$ 1.25
GPS _{GCN} +Performer (RWSE)	84.72 $\pm$ 0.65	48.08 $\pm$ 0.85	92.88 $\pm$ 0.50	84.81 $\pm$ 0.86	76.45 $\pm$ 1.51
GPS _{GCN} +Performer (DEG)	83.38 $\pm$ 0.68	48.93 $\pm$ 0.47	93.60 $\pm$ 0.47	80.49 $\pm$ 0.97	74.24 $\pm$ 1.18
GPS _{GAT} +Performer (LapPE)	85.93 $\pm$ 0.52	48.86 $\pm$ 0.38	92.62 $\pm$ 0.79	84.62 $\pm$ 0.54	76.71 $\pm$ 0.98
GPS _{GAT} +Performer (RWSE)	87.04 $\pm$ 0.58	49.92 $\pm$ 0.68	91.08 $\pm$ 0.58	84.38 $\pm$ 0.91	77.14 $\pm$ 1.49
GPS _{GAT} +Performer (DEG)	85.54 $\pm$ 0.58	51.03 $\pm$ 0.60	91.52 $\pm$ 0.46	82.45 $\pm$ 0.89	76.51 $\pm$ 1.19
GPS _{GCN} +Transformer (LapPE)	OOM	OOM	91.82 $\pm$ 0.41	83.51 $\pm$ 0.93	OOM
GPS _{GCN} +Transformer (RWSE)	OOM	OOM	91.17 $\pm$ 0.51	83.53 $\pm$ 1.06	OOM
GPS _{GCN} +Transformer (DEG)	OOM	OOM	91.76 $\pm$ 0.61	80.82 $\pm$ 0.95	OOM
GPS _{GAT} +Transformer (LapPE)	OOM	OOM	92.29 $\pm$ 0.61	84.70 $\pm$ 0.56	OOM
GPS _{GAT} +Transformer (RWSE)	OOM	OOM	90.82 $\pm$ 0.56	84.01 $\pm$ 0.96	OOM
GPS _{GAT} +Transformer (DEG)	OOM	OOM	91.58 $\pm$ 0.56	81.89 $\pm$ 0.85	OOM
GT	86.51 $\pm$ 0.73	51.17 $\pm$ 0.66	91.85 $\pm$ 0.76	83.23 $\pm$ 0.64	77.95 $\pm$ 0.68
GT-sep	87.32 $\pm$ 0.39	52.18 $\pm$ 0.80	92.29 $\pm$ 0.47	82.52 $\pm$ 0.92	78.05 $\pm$ 0.93
Heterophily-Designated GNNs					
CPGNN	63.96 $\pm$ 0.62	39.79 $\pm$ 0.77	52.03 $\pm$ 5.46	73.36 $\pm$ 1.01	65.96 $\pm$ 1.95
FAGCN	65.22 $\pm$ 0.56	44.12 $\pm$ 0.30	88.17 $\pm$ 0.73	77.75 $\pm$ 1.05	77.24 $\pm$ 1.26
FSGNN	79.92 $\pm$ 0.56	52.74 $\pm$ 0.83	90.08 $\pm$ 0.70	82.76 $\pm$ 0.61	78.86 $\pm$ 0.92
GBK-GNN	74.57 $\pm$ 0.47	45.98 $\pm$ 0.71	90.85 $\pm$ 0.58	81.01 $\pm$ 0.67	74.47 $\pm$ 0.86
GloGNN	59.63 $\pm$ 0.69	36.89 $\pm$ 0.14	51.08 $\pm$ 1.23	73.39 $\pm$ 1.17	65.74 $\pm$ 1.19
GPR-GNN	64.85 $\pm$ 0.27	44.88 $\pm$ 0.34	86.24 $\pm$ 0.61	72.94 $\pm$ 0.97	55.48 $\pm$ 0.91
H2GCN	60.11 $\pm$ 0.52	36.47 $\pm$ 0.23	89.71 $\pm$ 0.31	73.35 $\pm$ 1.01	63.59 $\pm$ 1.46
JacobiConv	71.14 $\pm$ 0.42	43.55 $\pm$ 0.48	89.66 $\pm$ 0.40	68.66 $\pm$ 0.65	73.88 $\pm$ 1.16
Graph SSMs					
GMN	87.69 $\pm$ 0.50	54.07 $\pm$ 0.31	91.01 $\pm$ 0.23	84.52 $\pm$ 0.21	—
GPS + Mamba	83.10 $\pm$ 0.28	45.13 $\pm$ 0.97	89.93 $\pm$ 0.54	83.70 $\pm$ 1.05	—
Ours					
TANGO _{GCN}	89.67 $\pm$ 0.68	52.98 $\pm$ 0.71	98.37 $\pm$ 0.49	85.57 $\pm$ 0.73	79.86 $\pm$ 1.14
TANGO _{GIN}	89.19 $\pm$ 0.62	50.76 $\pm$ 0.47	97.38 $\pm$ 0.50	84.39 $\pm$ 0.61	78.84 $\pm$ 0.96
TANGO _{GatedGCN}	91.89 $\pm$ 0.30	52.60 $\pm$ 0.53	98.32 $\pm$ 0.59	85.51 $\pm$ 0.98	80.39 $\pm$ 1.04
TANGO _{GPS}	91.08 $\pm$ 0.57	53.83 $\pm$ 0.32	98.39 $\pm$ 0.54	85.66 $\pm$ 1.01	80.32 $\pm$ 1.07